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## Systematic Literature Review of Social Media Bots Detection Systems

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### ABSTRACT

Online social networks (OSNs) are vital to people's daily lives. They offer free services that allow people to connect and interact with family and friends, post comments and images, express views on sports and politics, and influence other users on OSN. The significant risks to OSN security are malicious SMBs, and numerous studies have been done to identify them. This article, a Systematic Literature Review (SLR), aims to determine best practices in SMB recognition. This SLR covers research published between 2008 and 2022. As a result of this study, we classified OSN profiles into real, verified, and fake accounts. The malicious SMBs types are SMBs, spam bots, Sybil and cyborgs, stegobots, political bots, and game bots. We also proposed a classification of SMBs detection techniques, which are ML-based, DL-based, graph-based, anomaly-based, topic-modeling-based, DNA-inspired, genetic-based, and hybrid approaches. Additionally, our study revealed that most public datasets used for SMB detection only include some types of SMB, dependent on Twittersphere, include limited data, and needed to be more extensive, up-to-date and accurate. We studied challenges of SMB detection, like data labeling, features engineering, imbalanced data, etc. Finally, we proposed the opening axis for future works.

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## 1. Introduction

The emergence of OSNs, including Twitter, Facebook, and Instagram, has occurred in online communication. OSNs are now an essential component of communication for individuals, institutions, organizations, and businesses to stay in touch with friends and family and to maintain business relationships with clients and customers. According to "<https://www.statista.com>", Facebook continues to be the largest and most popular OSN in history, with over 2895 million (M) members, followed by YouTube (2514 M), WhatsApp (2000 M), Instagram (2000 M), TikTok (1051 M), Facebook Messenger (931 M), and Twitter (556 M). Parallel to that, these networks have evolved into performing additional tasks overseen by an autonomous agent known as a bot. A bot is a software that mimics human behavior and operates autonomously and automatically. His actions can be used for good or bad. An SMB is a social media bot that works. "Statista.com" estimates that 71% of Twitter accounts discussing U.S. stocks are social bots, whereas Twitter has only identified 37% of them. A recent study (Cresci et al., 2018) exposed the actions of spam bots in Twitter stock microblogs. The research revealed that 71% of Twitter accounts that retweeted stock posts were bots, and 37% of these accounts had their accounts banned. 2019 saw the identification of 11% of Facebook accounts as bots (Zago, et al., 2019), and a study found that social bots contributed to the "infodemics" by disseminating false information on the COVID-19 epidemic (Gallotti et al., 2020).

Bots can operate within networks, which are referred to as OSN botnets. It is a group of SMBs that collaborate to perform specific actions on OSN and are generated and remotely managed by a bot master via a command and control (C&C) channel. To avoid detection by bot detection systems, social botnets mimic the interactions of OSN users. (Ellaky et al., 2021). The SMB's creation is effortless today by utilizing automation and bot-building tools like "Seek Socially" and "UseViral". The most developed bots use cloud services like Heroku or Amazon Web Services (AWS). Additionally, cloud service companies offer Bot-as-a-Service (BaaS) services, which have intelligence features like producing sophisticated content or disseminating spam content. An SMB comprises two main components: the program that manages the account and the exposed account on OSN. An SMB usually allows two different types of generic operations. The first is social engagement, such as reading, writing, and responding to information uploaded by other users. The second is social networking capabilities like adding friends or following other users that are used to change the social network. Social media can be used for various purposes, including brand marketing, teamwork, networking, and communi-

cation. People and organizations use SMBs to increase their effect, distribute ideas and materials, and widen their reach. Social bots are capable of both malicious and good deeds (Orabi et al., 2020). Real accounts and verified profiles owned by companies, celebrities, and other entities manage legal operations. They were created to supervise business operations or to assist in the relationship between a client and a company, including exchanging goods and services, publishing earthquake alerts on Twitter, disseminating news, etc. On the other hand, malicious bots can also harass or burden users by engaging in unethical activities, such as stealing the identities of actual users, swaying the electorate to support specific candidates, and propagating hate speech and other polarizing content. The use of social bots in OSNs has been investigated in several fields, including politics (Ferrara, 2017); (Luceri et al., 2019), stock market manipulation (Ferrara, 2015), fake news (Grinberg et al., 2016), the propagation of extremist ideologies and hate speech (Shao et al., 2018); (Stella et al., 2018), health misinformation propaganda involving vaccines (Sutton, 2018); (Broniatowski, et al., 2018).

Comprehending, detecting, and categorizing social bots and botnets has taken much work. Therefore, the development of precise and scalable bot detection systems has been facilitated by breakthroughs in artificial intelligence, particularly those in machine learning (ML) and deep learning (DL) approaches. Even though there are more publications every year to stay up with social bot evolution, these latter continue to pose new risks and difficulties for detection systems.

This study aims to conduct an SLR of social bot detection technologies. To the best of our knowledge, only a few state-of-the-art articles (Orabi et al., 2020); (Alothali et al., 2018); (Ferrara et al., 2016); (Alothali et al., 2018) have addressed social bot identification. We acknowledge the value of the existing reviews. Still, we highlight that they were conducted a few years ago and may need to capture the most recent developments in social bot detection. It is crucial to conduct a review on social bot detection despite the existing reviews because the field of SMB detection is constantly evolving, and new methods and techniques are being developed. This SLR covers research on social bot detection techniques published between 2008 and 2022 and addresses various social bot kinds, including sybils, cyborgs, spam bots, stegobots, and political bots, on OSNs including Facebook, Twitter, Instagram, YouTube, Renren, and Sina Weibo. This SLR study offers a concise and clear summary of the methods currently used to categorize various types of SMBs, suggests doing a quantitative and analytical analysis, and gives future research directions. Furthermore, in our work, we studied all of a model's phases and life cycle, from data preparation to feature engineering to feature

selection to dataset labeling, etc., which was not done in the previous state of the art.

The significant contributions of this research work are as follows:

- 1) Review of the most recent research works produced to detect SMBs.
- 2) Classification and categorization of detection techniques.
- 3) Identification of data sources, datasets, feature types, feature engineering techniques, detection approaches, algorithms, and performance evaluation.
- 4) Analysis of detection techniques for each type of SMB.

This article is presented as follows. Section II introduces the research methodology of our SLR. Next, Section III provides a literature review of works related to SMB detection. Section IV provides a complete overview of the research's findings. Section V gives a general discussion of the results. Finally, Section VI provides the conclusion and prospects.

## 2. Research methodology

A SLR is a type of literature review in which a collection of articles on a specific topic is gathered and critically analyzed in order to identify, select, synthesize, and assess relationships, limitations, and key findings in order to summarize both quantitative and qualitative studies. (Xiao and Watson, 2019). The SLR is a crucial step for researchers to investigate the current research topics regarding the detection technique of SMBs, highlight the weaknesses of the proposed approaches, and give new trends and orientations for researchers. We used the PRISMA approach, a widely recognized and rigorous method for conducting SLRs. It involves a structured approach to searching for and selecting relevant literature, as well as assessing the quality of the studies included in the review. The PRISMA approach consists of several key steps, including: Defining the research question and scope of the review; conducting a comprehensive search of relevant databases and other sources of literature; screening the search results based on predefined inclusion and exclusion criteria; extracting data from the selected studies and synthesizing the findings; and assessing the quality of the studies included in the review.

### 2.1. Search strategy

We used the most popular research databases because the search strategy directly impacts the relevance and completeness of the studies retrieved: ScienceDirect, the IEEE Xplore digital library, SpringerLink, ACM, ResearchGate, arXiv, and Google Scholar.

**Table 1**  
Inclusion and exclusion criteria.

| Inclusion Criteria                             | Exclusion Criteria                                                          |
|------------------------------------------------|-----------------------------------------------------------------------------|
| Paper-indexed Scopus or Web of Science.        | Studies that are not related to computer science or information technology. |
| The paper is redacted in English.              | Articles on social bots that do not propose detection systems.              |
| The articles are from journals or conferences. | Absence of Digital Object Identifier (DOI).                                 |

### 2.2. Research questions

To understand the problem of social bot detection in OSNs and its challenges and to highlight the different aspects of this problem, we established the questions that this SLR must address as follows:

**RQ1:** What are the existing types of OSN profiles and SMBs? This question will help determine the different accounts and bots operating on OSNs.

**RQ2:** What are the approaches used to detect SMBs? This question aims to determine the methods used to detect and classify SMBs and to study their performance.

**RQ3:** Which features are used to identify SMBs effectively, and which are the most relevant? This question determines the account characteristics used in the detection process and shows the most decisive ones.

**RQ4:** Which feature engineering and preprocessing techniques are used for social bot detection? This question identifies the preprocessing techniques applied to data to improve their quality.

**RQ5:** What are the dataset's characteristics used for social bot detection? This question aims to identify the sources of data, the different problems related to datasets, and how data can be labeled.

### 2.3. Query terms

We used snowballing method, which involves starting with a small set of identified studies, often the most relevant or seminal papers in a field, and then expanding the search by identifying and examining the references cited in those papers. The process is repeated, with each new set of papers examined for additional references to include in the review. This iterative process continues until no further relevant studies are identified. The steps conducted for our snowballing research are described as follows:

- Employ a text mining process to find search keywords and terms pertinent to SMBs detection.
- Conduct a broad initial search using a search string constructed from what academics deem pertinent search terms.
- Indicate whether the resulting studies are pertinent (P) or not (NP) for the current review.
- Use text mining techniques on both P and NP texts to uncover and extract new key phrases and identify old essential terms that are no longer relevant.
- Update the search term and go back to step one.
- Then, using Boolean operators, a search process was carried out by ScienceDirect, the IEEE Xplore digital library, SpringerLink, ACM, ResearchGate, arXiv, and Google Scholar to find the pertinent articles that addressed the search questions based on pre-determined keywords.

The aforementioned search methodology and screens have been merged with additional phrases based on the knowledge discovered during the searches to construct a suitable search string for the SLR now under development. So, the last search term we used for the search queries was:

**Query 1:** ("Social bots" OR "Social media bot\*" OR "OSN bots" OR "digital platforms bot\*").

**Query 2:** ("Sybil" OR "Spam bots" OR "Spammer\* Social network" OR "Stegobot" OR "Cyborg" OR "Fake accounts" OR "Malicious social media bots").

**Query 3:** ("Automated account\*" OR "Trending Topics" OR "generating posts" OR "like farms" OR "automated repost\*").

**Query 4:** ("Social bots" detection" OR "Technique\* to detect social bots" OR "Detection of social media bot\*").

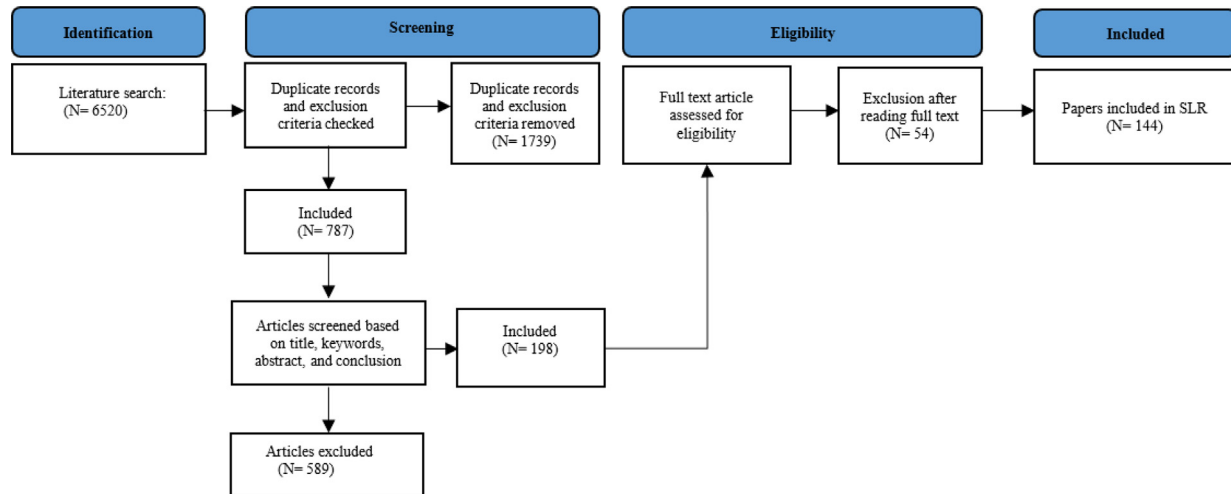


Fig. 1. Flowchart of the search process.

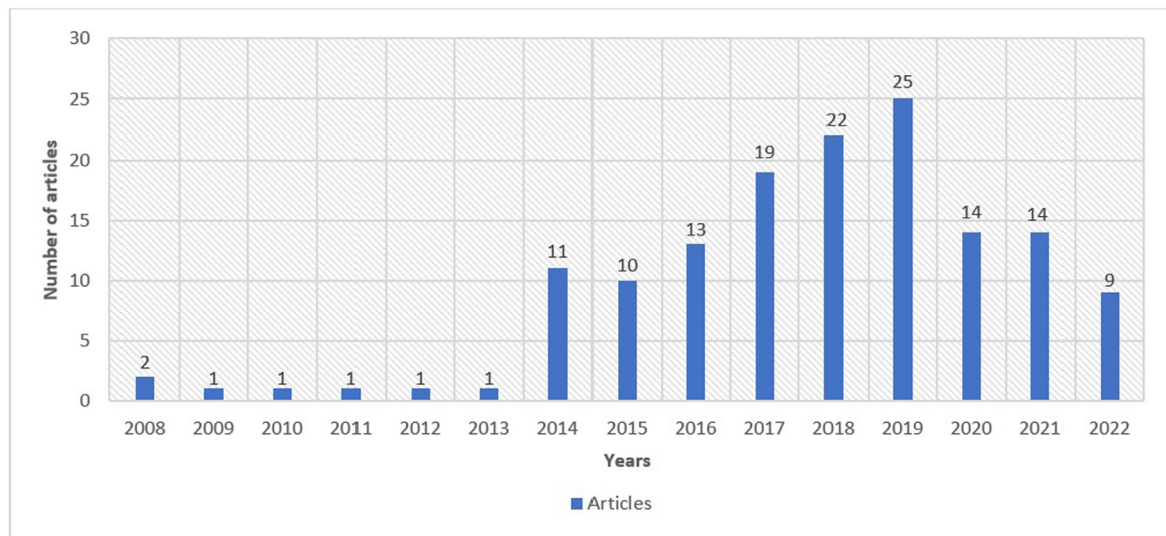


Fig. 2. Distribution of the state-of-the-art works over the years.

#### 2.4. Study selection and quality assessment

The paper selection step was conducted based on a list of inclusion and exclusion criteria to ensure the pertinence and quality of papers. Table 1 presents the requirement of the article to be included in the SLR in column one and the exclusion criteria in the second column.

Specific quality assessment guidelines are used to determine the most pertinent works to evaluate the articles' quality. The first parts of the selected papers assessed were the title, abstract, conclusion, and keywords. Articles that do not correspond with our study were eliminated. The remaining articles were then subjected to the quality evaluation criteria for their dependability, integrity, and relevancy. For both systematic reviews and research syntheses, we developed a checklist of 12 guidelines with this objective in mind. If the work partially complies with the rule is merited in each situation:

- Satisfy the rules score = 1
- Satisfy some of the rules score = 0.5
- Does not satisfy the rule score = 0

Then The ultimate score of the article is determined by adding up the points awarded for each of the twelve quality evaluation criteria. As a result, only articles with a rating of 8 to 14 are included in the selected papers.

The rules are defined as follows:

1. Do the chosen articles refer to the research query?
2. Is the paper well-structured?
3. Do the research aims in the literature appear appropriately stated?
4. Does the literature provide sufficient baseline knowledge regarding SMBs?
5. Does the literature explain the terminology they used?

**Table 2**  
Publication venues.

| Document Type       | Source Title                                                                                                                                     | References                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Conference Articles | International Conference on Digital Age and Technological Advances for Sustainable Development                                                   | (Ellaky et al., 2021)                                                                       |
|                     | AAAI Conference on Artificial Intelligence                                                                                                       | (Yang et al., 2020)                                                                         |
|                     | ACM Asia Conference on Computer and Communications Security                                                                                      | (Lingam et al., 2020)                                                                       |
|                     | ACM Conference on Computer and Communications Security                                                                                           | (Song et al., 2015)                                                                         |
|                     | ACM Conference on Web Science                                                                                                                    | (Cresci et al., juin 2019) (Mazza et al., 2019)                                             |
|                     | ACM International Conference on Information & Knowledge Management                                                                               | (Feng et al., 2021) (Feng et al., 2021) (Sayyadiharikandeh et al., 2020) (Cai et al., 2017) |
|                     | Multidisciplinary International Social Networks Conference                                                                                       | (El-Mawass et al., 2018)                                                                    |
|                     | ACM Transactions on Privacy and Security                                                                                                         | (Ikram, et al., 2017)                                                                       |
|                     | Artificial Intelligence Methods in Intelligent Algorithms                                                                                        | (Daouadi et al., 2019)                                                                      |
|                     | Australasian Conference on Information Systems                                                                                                   | (Stieglitz et al., 2017)                                                                    |
|                     | Applications and Techniques in Information Security                                                                                              | (Ma et al., 2014)                                                                           |
|                     | Bio-Inspired Computing - Theories and Applications                                                                                               | (Yang, et al., 2014)                                                                        |
|                     | Companion of the World Wide Web Conference                                                                                                       | (Davis et al., 2016) (Zannettou et al., 2019) (Ali Alhosseini et al., 2019)                 |
|                     | Communications of the ACM                                                                                                                        | (Ferrara et al., 2016)                                                                      |
|                     | European Network Intelligence Conference                                                                                                         | (Teljstedt et al., 2015)                                                                    |
|                     | IEEE Conference on Communications and Network Security                                                                                           | (Feng et al., juin 2019)                                                                    |
|                     | IEEE International Conference on Big Data                                                                                                        | (Khaled et al., 2018)                                                                       |
|                     | IEEE International Conference on Data Mining                                                                                                     | (Chavoshi et al., 2016)                                                                     |
|                     | IEEE International Conference on Intelligence and Security Informatics                                                                           | (Fernquist et al., 2018)                                                                    |
|                     | IEEE International Conference on Communications                                                                                                  | (Chen et al., 2015)                                                                         |
|                     | IEEE Symposium on Security and Privacy                                                                                                           | (Yu et al., 2008)                                                                           |
|                     | IEEE Transactions on Information Forensics and Security                                                                                          | (Dutta et al., 2020)                                                                        |
|                     | IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining                                                             | (Dickerson et al., 2014) (Morstatter et al., 2016) (Minnich et al., 2017)                   |
|                     | IFIP Advances in Information and Communication Technology                                                                                        | (Feng et al., 2021) (Beskow and Carley, 2018) (Liu et al., 2014)                            |
|                     | Information Retrieval Technology                                                                                                                 | (Mulamba et al., 2016)                                                                      |
|                     | Proceedings of the Fourth International Workshop on Social Sensing                                                                               | (Almerekhi and Elsayed, 2015)                                                               |
|                     | International AAAI Conference on Web and Social Media                                                                                            | (Hurtado et al., 2019)                                                                      |
|                     | International Bhurban Conference on Applied Sciences and Technology                                                                              | (Cresci et al., 2018)                                                                       |
|                     | International Conference for Convergence in Technology                                                                                           | (Mateen et al., 2017)                                                                       |
|                     | International Conference on Social Informatics                                                                                                   | (Kiran et al., 2019)                                                                        |
|                     | International Conference on Communication Technology Proceedings                                                                                 | (Chen et al., 2017)                                                                         |
|                     | International Conference on Computer Science and Engineering                                                                                     | (H. Ping et al., 2018)                                                                      |
|                     | International Conference on Computer, Communication, and Signal Processing                                                                       | (M. Kantepe et al., 2017)                                                                   |
|                     | International Conference on Data Intelligence and Security                                                                                       | (Kumar et al., mai 2021)                                                                    |
|                     | International Conference on Data Science and Advanced Analytics                                                                                  | (Shaabani et al., 2018)                                                                     |
|                     | International Conference on Dependable, Autonomic, and Secure Computing                                                                          | (Pizarro, 2020)                                                                             |
|                     | International Conference on Pervasive Intelligence and Computing                                                                                 | (Dorri et al., 2018)                                                                        |
|                     | International Conference on Big Data Intelligence and Computing                                                                                  |                                                                                             |
|                     | IEEE Cyber Science and Technology Congress                                                                                                       |                                                                                             |
|                     | International Conference on Industrial and Information Systems                                                                                   | (Lingam et al., 2018)                                                                       |
|                     | International Conference on Intelligent Systems Design and Applications                                                                          | (De Meo et al., 2011)                                                                       |
|                     | IEEE Conference on Computer Communications                                                                                                       | (Wang et al., 2017)                                                                         |
|                     | International Conference on Innovations in Information Technology                                                                                | (Alothali et al., 2018)                                                                     |
|                     | International Conference on Social Networks Analysis, Management, and Security                                                                   | (Bello et al., 2018)                                                                        |
|                     | International Conference on Intelligent Computing in Data Sciences                                                                               | (Ellaky et al., 2021)                                                                       |
|                     | International Conference on Web Intelligence, Mining, and Semantics                                                                              | (Beğenilmiş and Uskudarli, 2018)                                                            |
|                     | International Conference on Machine Learning and Applications                                                                                    | (Alsaleh et al., 2014)                                                                      |
|                     | International Conference on Services Systems and Services Management                                                                             | (Jin Dan and Teng Jieqi, 2017)                                                              |
|                     | International Conference on ICT Systems Security and Privacy Protection                                                                          | (Mulamba et al., 2016)                                                                      |
|                     | International Conference on Trust, Privacy, and Security in Intelligent Systems and Applications                                                 | (Wei and Nguyen, 2019)                                                                      |
|                     | International Conference on Web and Social Media                                                                                                 | (Varol et al., 2017)                                                                        |
|                     | International Workshop on Information Forensics and Security                                                                                     | (Andriotis and Takasu, 2018)                                                                |
|                     | International Conference on World Wide Web Companion                                                                                             | (Cresci et al., 2017)                                                                       |
|                     | International Conference on Social Computing, Behavioral-Cultural Modeling and Prediction and Behavior Representation in Modeling and Simulation | (Chew et al., 2018)                                                                         |
|                     | International Conference on Human-Computer Interaction                                                                                           | (Castillo, et al., 2017)                                                                    |
|                     | Procedia Computer Science                                                                                                                        | (Fernandes et al., 2015) (Pratama and Rakhmawati, 2019)                                     |
|                     | Proceedings of the World Wide Web Conference                                                                                                     | (Abu-El-Rub and Mueen, 2019)                                                                |
|                     | Proceedings of the Annual Computer Security Applications Conference                                                                              | (Chu et al., 2010)                                                                          |
|                     | Proceedings of the 9th USENIX conference on Networked Systems Design and Implementation                                                          | (Cao et al., 2012)                                                                          |
|                     | Proceedings of the Network and Distributed System Security Symposium                                                                             | (Danezis and Mittal, 2009)                                                                  |
|                     | World Conference on Smart Trends in Systems, Security, and Sustainability                                                                        | (Mohammad et al., 2019)                                                                     |
|                     | World Symposium on Web Applications and Networking                                                                                               | (AlRubaian et al., 2015)                                                                    |

(continued on next page)



Table 2 (continued)

| Document Type    | Source Title                                                                    | References                                                                                                                 |
|------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Journal Articles | American Journal of Public Health                                               | (Sutton, 2018) (Broniatowski, et al., 2018)                                                                                |
|                  | ACM Transactions on Knowledge Discovery from Data                               | (Costa et al., nov. 2017) (Yang et al., févr. 2014)                                                                        |
|                  | ACM Transactions on Multimedia Computing, Communications, and Applications      | (Jr et al., avr. 2018)                                                                                                     |
|                  | ACM Transactions on the Web                                                     | (Gilani et al., févr. 2019)                                                                                                |
|                  | ACM SIGWEB Newsletter                                                           | (Ferrara, 2015)                                                                                                            |
|                  | Applied Network Science                                                         | (Luceri et al., 2019)                                                                                                      |
|                  | Applied Sciences (Switzerland)                                                  | (Zhao et al., 2020)                                                                                                        |
|                  | Automatic Control and Computer Sciences                                         | (Zegzhda et al., 2017)                                                                                                     |
|                  | Computational and Mathematical Organization Theory                              | (Beskow and Carley, mars 2019)                                                                                             |
|                  | Computer Communications                                                         | (Lee and Kim, 2014)                                                                                                        |
|                  | Computer Journal                                                                | (Natarajan et al., 2015)                                                                                                   |
|                  | Cluster Computing                                                               | (Murugan and Devi, 2019)                                                                                                   |
|                  | Computers and Security                                                          | (Ji et al., 2016) (García et al., sept. 2014) (Rodríguez-Ruiz et al., 2020)                                                |
|                  | Decision Support Systems                                                        | (Boshmaf, août 2016) (Liu et al., août 2017) (Akiyama et al., août 2017)                                                   |
|                  | EPJ Data Science                                                                | (Cresci et al., déc. 2015)                                                                                                 |
|                  | First Monday                                                                    | (Bruno et al., 2022)                                                                                                       |
|                  | Future Generation Computer Systems                                              | (Ferrara, 2017)                                                                                                            |
|                  | Human Behavior and Emerging Technologies                                        | (Al-Qurishi et al., 2018)                                                                                                  |
|                  | IEEE Access                                                                     | (Yang et al., 2019)                                                                                                        |
|                  | IEEE/ACM Transactions on Networking                                             | (Martin-Gutierrez et al., 2021) (Wu et al., 2020) (Loyola-Gonzalez et al., 2019) (M. S. Al-Rakhami et A. M. Al-Amri, 2020) |
|                  | IEEE Communications Magazine                                                    | (Yu et al., 2008)                                                                                                          |
|                  | IEEE Intelligent Systems                                                        | (Zago, et al., 2019)                                                                                                       |
|                  | IEEE Systems Journal                                                            | (Cresci et al., 2016)                                                                                                      |
|                  | IEEE Transactions on Computational Social Systems                               | (Lingam et al., 2021)                                                                                                      |
|                  | IEEE Transactions on Control of Network Systems                                 | (Rout et al., 2020)                                                                                                        |
|                  | IEEE Transactions on Industrial Informatics                                     | (Wang and Paschalidis, 2017)                                                                                               |
|                  | IEEE Transactions on Dependable and Secure Computing                            | (Al-Qurishi et al., 2018)                                                                                                  |
|                  | Information Processing and Management                                           | (Yang et al., 2016) (Cresci et al., 2017) (Chu et al., 2012)                                                               |
|                  | Information Sciences                                                            | (Orabi et al., 2020)                                                                                                       |
|                  | Information Systems                                                             | (Kudugunta and Ferrara, 2018) (Igawa, et al., 2016) (Miller et al., 2014)                                                  |
|                  | Inteligencia Artificial                                                         | (Jeong et al., 2016)                                                                                                       |
|                  | International Journal of Electrical and Computer Engineering                    | (Pham et al., 2022)                                                                                                        |
|                  | Journal of Big Data                                                             | (Safaa et al., 2022)                                                                                                       |
|                  | Journal of Planning education and Research                                      | (Benabbou et al., 2022)                                                                                                    |
|                  | Journal of Cyber Security and Mobility                                          | (Koggalahewa et al., 2022)                                                                                                 |
|                  | Journal of Information Security and Applications                                | (Xiao and Watson, 2019)                                                                                                    |
|                  | Journal of Computational Social Science                                         | (Shetty et al., 2021)                                                                                                      |
|                  | Knowledge-Based Systems                                                         | (Wanda and Jie, 2020)                                                                                                      |
|                  | Multimedia Tools and Applications                                               | (Yang et al., 2022)                                                                                                        |
|                  | Nature Communications                                                           | (Wu et al., 2021)                                                                                                          |
|                  | Nature Human Behaviour                                                          | (Venkatachalam and Anitha, 2017)                                                                                           |
|                  | Neurocomputing                                                                  | (Shao et al., 2018)                                                                                                        |
|                  | Online Social Networks and Media                                                | (Gallotti et al., 2020)                                                                                                    |
|                  | PLoS ONE                                                                        | (Inuwa-Dutse et al., 2018) (Wu et al., 2016)                                                                               |
|                  | Proceedings of the National Academy of Sciences of the United States of America | (Cresci et al., 2019)                                                                                                      |
|                  | Science                                                                         | (Rauchfleisch and Kaiser, 2020)                                                                                            |
|                  | Social Network Analysis and Mining                                              | (Stella et al., 2018)                                                                                                      |
|                  | Social and Information Networks                                                 | (Grinberg et al., 2016)                                                                                                    |
|                  | Social Science Computer Review                                                  | (Wanda and Jie, 2021) (Alothali et al., 2021)                                                                              |
|                  | SpringerPlus                                                                    | (Gao et al., 2018) (Kharaji and Rizi, 2014) (Balestrucci et al., 2021)                                                     |
|                  |                                                                                 | (Hagen et al., 2022)                                                                                                       |
|                  |                                                                                 | (Kang et al., 2016)                                                                                                        |

6. Are the detected SMBs types precisely defined?
7. Does the literature mention the source of the data used?
8. Does the literature mention the size and characteristics of the used datasets?
9. Is the detection experiment that was done appropriate and acceptable?
10. Is the research that was done on the ground legitimate?
11. Can you readily identify the dataset used?
12. Are the detection experiment results identified and published in detail?
13. Are the techniques utilized to evaluate the outcomes of the detection experiments suitable?
14. Is the method suggested by the literature considered valuable?

## 2.5. Study selection and flowchart of search process

Our study selection and research process followed four stages. In the first stage, 6420 papers were selected and got through the SLR process. At the screening stage, we excluded duplicate papers, applied exclusion criteria, and kept the papers on the research topic based on title, abstract, keywords, and findings. The unrelated articles were eliminated from the review. The quality assessment criteria were then used to evaluate the rest of the articles for trustworthiness, integrity, and relevance. Only 144 papers were identified as relevant. Fig. 1 represents the flowchart conducted for our search process.

## 2.6. Data analysis

We identified 144 studies published in the field of social bot detection during the period between 2008 and 2022. Fig. 2 displays the distribution of the state-of-the-art articles by year.

So, we can see that most of the investigations on social bots were conducted between 2017 and 2019. Research studies and publications related to this concept also increased significantly in 2014. The number of research work about social media bots has risen in recent years due to the proliferation of social media platforms and their impact on politics. Bots are often used to spread misinformation or propaganda, and their ability to amplify messages quickly and widely has raised concerns about their influence on public opinion. Researchers are interested in understanding how bots influence conversations on social media platforms and developing methods for detecting and preventing their use. Overall, the increasing use of social media and the impact of bots on public opinion and politics have made them an important area of research in recent years. Additionally, advancements in technology, like ML, DL, and NLP techniques and data availability, have made it easier and more feasible to conduct research on social media detection in 2019. Table 2 presents the publication venues of the retrieved articles, and the result shows that 50% are conference articles and 50% are journal articles.

## 2.7. Data extraction strategy

Once the cutting-edge papers have passed through the preceding stages, additional information is retrieved from the article based on the research objectives and study questions to thoroughly understand research issues and each work's contribution to social bot detection. The most important extracted information is the reference, article title, authors' names, publication year, type of detection method (classification or clustering or hybrid, etc.), type of SMB, OSN platform, dataset, labeling techniques, feature categories, feature extraction techniques, feature selection techniques, pre-processing techniques, models and algorithms, optimization techniques, performance metrics, and the state-of-the-art techniques compared to. The information gathered serves as the foundation for this study. The following sections will present and discuss it using graphs, figures, and comparative tables to conduct a methodical investigation to help readers understand SMBs and their detection challenges.

## 3. Related work

The scope of this section is to give the readers a summary of different proposals for SMBs detection techniques. In several research, the goal was to detect whether the OSN user is a human or a bot (normal /social bots, malicious/legitimate) without specifying the type of bot. In this case, the paper aims to detect a generic SMB. The other research papers specify the type of bot targeted, such as Sybil and cyborgs, Stegobot, Game bots, political bots, and spam bots. As a result, our literature review organizes detection techniques by the type of SMB studied: SMB in general, Sybil and Cyborg, Stegobot, Political bot, and Game bot.

### 3.1. Generic SMBs detection techniques

The authors of this paper (Gilani et al., 2019) scraped a massive dataset from Twitter that included user information and content features. Then proceeded, an analysis of behavioral features such as the number of likes per tweet, the number of retweets per tweet, the number of replies, the mentions, etc. The findings of this study showed that Twitter bots tweet and re-tweet more than humans,

post five times more URLs in their tweets than human posts, and receive more likes to their content.

### • ML-based detection systems

The authors in (Dickerson et al., 2014) proposed a social bot detection system based on the AdaBoost model. They used network features, including a subgraph of the Twitter follower network of 40 M edges, content, and sentiment features. Several ML algorithms have been applied to detect SMBs, and the AdaBoost model gave the best AUC score of 73%. In addition, the authors showed that sentiment features are essential to identify bots. In (Lee and Kim, 2014), a new detection scheme was proposed to identify groups of malicious users once created. This method is employed to discrepancy between automatically produced usernames and ones created by effective human users. A clustering strategy was applied to group the accounts with similar names, and Support Vector Machine (SVM) algorithm categorized the groups as malicious or not. The study analyzed approximately 4.7 million Twitter accounts collected using an API. In (Yang, et al., 2014), the authors presented a new bot detection technique using an interaction graph based on user interaction features and a Back Propagation Neural Network algorithm (BPNN) model. BPNN algorithm with the interaction features performed an accuracy of 98.6% and outperformed the existing techniques. BotOrNot (Davis et al., 2016) is a system that evaluates the level of the menace of a Twitter user. Therefore, the data were extracted from the user profiles and classified into the following feature categories: network, user, friends, temporal, content, and sentiment features. BotOrNot is based on Random Forest (RF), the machine learning algorithm which gave the best AUC rate of 95%. A host-side system (Ji et al., 2016) can identify social botnets. Based on the analysis of evasion strategies employed by social botnets, the authors proposed new features classified as life cycle features, temporal, and spatial correlations. The RF model gave good results compared to the other ML algorithms, with a False Positive (FP) of 0.3%, a False Negative (FN) of 4.7%, and an F-measure of 96.3%. This work (Ikram, et al., 2017) provided a honeypot-based detection system using SVM and the co-clustering method. For this aim, the authors extracted data from Facebook users' timelines and classified features into lexical and non-lexical classes using the SVM algorithm. The non-lexical features included comments, likes, and shared content words per post, and the lexical features included grammatical and semantic correctness of the posted content. As a result, the model achieved a precision of 99% and a recall of 93%. In this study (Kantepe and Ganiz, 2017), the authors proposed SMB system detection based on behavioral features and the Gradient boosting Trees algorithm (GBT), which achieved an accuracy of 86%. Therefore, they applied Information Gain (IG), Mutual Information (MI), and Chi-Square as feature selection techniques. In this paper (Jin Dan and Teng Jieqi, 2017), behavioral analysis was conducted based on RF, which reached a precision of 94.4% using data from Sina-Weibo. The analysis showed that user information, content, and behavioral features are the most relevant for bots' identification. The experiments confirmed that the suggested attributes positively impacted identifying the bots. SocialBotHunter (Dorri et al., 2018) is a semi-supervised collective classification based on the One-Class SVM algorithm. This approach integrated network features, user information, and content features to identify Twitter botnets. The results showed that SocialBotHunter successfully detects spam bots with a precision of 99.95 % and an acceptable recognition time. To detect SMBs (Jr et al., 2018), the authors implemented a wavelet-based technique that classified social accounts as humans, legitimate or malicious bots based on content features. For feature extraction, the authors created a vector by applying discrete wavelet transformation and a weighting scheme

called lexicon-based coefficient attenuation. The RF classifier was trained on datasets scraped using the Twitter API, and an accuracy score of 94.47% was achieved. To detect SMBs using their social names, the authors (Beskow and Carley, 2019) applied an SVM-based model and text-based content, user metadata, temporal, and network features. To extract textual features, they used the TF/IDF of a bigram, the number of lower-case words, the number of upper-case words, and the string entropy based on Shannon entropy. As a result, SVM performed an accuracy of 99.6% and a recall of 99.8 %. This paper (Fernandes et al., 2015) proposed a bot detection system based on an SVM classifier and a DBSCAN clustering model trained using content, user information, and network features. DBSCAN achieved an F1-score of 91% but struggled to nuance human and bot behavior. After that, SVM is applied to classify bot users into three classes: brands, celebrities, and promoters, with an F1-score of 74%. In this article (Ellaky et al., 2021), the authors studied the impact of feature presentation on SMB detection using Bert and Doc2Vec Word Embedding. They applied RF and DT algorithms as classifiers, and the experiment results showed a precision of 99.96% for RF and DT with Bert and outperformed a precision of 100% with Doc2Vec. This article (Allothali et al., 2021) presented a hybrid feature selection technique for SMB detection based on RF algorithms and user information features. Four feature selection techniques were experimented with: wrappers, filters, embedded, and cross-validation attribute evaluation (CVAE) techniques. As a result, RF achieved an accuracy of 89% using CVAE. They also found that the best-performing features are favorites\_count, verified, statuses\_count, and average\_tweets\_per\_day. The authors of this article (Almerekhi and Elsayed, 2015) proposed a J48-based model for detecting Arabic social bots on Twitter using content and temporal features. The system was able to distinguish human accounts from bots with an accuracy of 92% using a dataset manually labeled with the help of the CrowdFlower platform. In this paper (Igawa, et al., 2016), the authors proposed a system for SMB classification using content features based on Lexicon Based Coefficient Attenuation (LBCA) and Wavelets. The RF model was trained on a dataset crawled from the 2014 FIFA World Cup, which performed an accuracy rate between 94% and 100%. This work (Rodríguez-Ruiz et al., 2020) presented an anomaly detection system implemented using RF as one-class ML classifiers and behavior features extracted from the Cresci-17 dataset. As a result, the proposed model achieved an AUC that exceeded 89% and allowed the detection of several types of bots. The authors of (Rout et al., 2020) proposed a social bot detection method based on content features. They added new features such as URL redirection, the relative position of the URL, frequency of shared URLs, spam content in the URL, anomalous URLs, etc. In addition, they evaluated the trustworthiness of tweets posted by each user using Bayesian learning and Dempster-Shafer theory (DST). Then they designed a learning automata-based malicious social bot detection LA-MSBD algorithm by incorporating a trust model with a set of URL-based features. As a result, the proposed system achieved a precision of 95.37% in detecting social bots. The authors in (Chen et al., 2017) proposed a system to detect active malicious botnets on Twitter by clustering tweets that contain embedded and short URLs. To train the model, they used real data extracted using the domain name of a URL shortening platform. Then picked a set of tweets that contained more than 20 duplicate tweets. The approach considered a cluster as a group of accounts sharing 200 tweets. In this study (Shaabani et al., 2018), the authors proposed an approach to detect social bots without using network features, user information, or content features. The suggested approach works based on collective causality metrics that provide a way to identify significant causal relationships. The results showed that the trained model, utilizing a

causality-based clustering, could recognize groups of SMBs with an accuracy of 75%. TwiBot-20 (Feng et al., 2021) is a massive benchmark for Twitter bots. It includes 229,573 accounts, 33.48 M tweets, 8.72 M user ownership items, and 455,958 following connections and includes all the features that could be scraped using the Twitter API. Authors compared different studies on this dataset and others as Cresci-17 and PAN-19. As a result, all the state-of-the-art detection systems fail to perform well on the TwiBot-20 dataset.

#### • Topic modeling based detection systems

The authors of (Morstatter et al., 2016) proposed a bot recognition model with an 82.48% recall using the AdaBoost algorithm. The study is based on content features from two real-world Twitter datasets: the Libya dataset, which included the Arab spring tweets in Libya, and the Arabic Honeypot dataset. The authors suggested new features, which are a fraction of the time of retweets, the mean time between tweets, the average length of tweets, number of URLs, and used the Latent Dirichlet allocation (LDA) for topic modeling. DeBot (Chavoshi et al., 2016) was developed to aggregate user accounts into a cluster in real time using only content features (comments) without needing a labeled dataset. The proposed clustering model used the warped correlation to detect thousands of SMBs per day with an accuracy of 94% and reported daily activities of the detected accounts. Based on a comparative study conducted by the authors of this work, they observed that certain bots escaped Twitter's suspension process and remained active for months.

#### • Anomaly-based systems

BotWalk (Minnich et al., 2017) is an anomaly-based system that identifies SMBs by detecting new behavior patterns using user information, content, temporal, and network features. BotWalk can detect new bots by performing anomaly detection using a clustering algorithm based on behavioral characteristics. Twitter accounts previously labeled as bots are randomly selected, and the follower's attributes of each account are extracted in real-time using the Twitter API and Tweepy. When new bots are detected, they are added to the seed bank of collected bots. Therefore, this system recognized 6000 candidate bots per day with a precision of 90%. In this paper (Costa et al., 2017), an activity model (ActM) was developed to accurately fit the distribution of OSN of inter-arrival times of a user's posts (IATs). The suggested model is an anomaly-based model built using an ActM. This mathematic-based algorithm uses a stochastic process to detect SMBs based on user posts in the timeline with an accuracy of 93%.

#### • DL-based detection systems

In this paper (Cai et al., 2017), the authors proposed an SMBs detection approach based on Convolutional Neural Network (CNN) model. The model learned the representation of user behavior by capturing the external and internal factors influencing the OSN user's actions. Then a Long Short-Term Memory (LSTM) algorithm was used to build a content-based representation as temporal data rather than just text. As a result, the model achieved an F-measure of 88.3%. In the study (Kudugunta and Ferrara, 2018), the authors proposed a system based on the LSTM algorithm and used content features and user information to recognize SMBs at the posting scale. Because of the need for a large dataset to train the DL model, the authors used oversampling to balance a labeled dataset. The experiments showed that the model could detect social bots from a single tweet with more than 96% accuracy. DeBD (Beğenilmiş and Uskudarli, 2018) is a



bot detection system based on the LSTM algorithm composed of three layers. The first one extracts the content features and the links between them. The second layer extracts the time-based features from the metadata of the tweets using the LSTM algorithm. The third layer mixed the extracted content features with the temporal features to identify the social bots with an accuracy of 100%. In this study (Mohammad et al., 2019), the authors implemented CNN and Artificial Neural Network (ANN) models to detect SMBs based on a single post. For the experiment, they used content features and user features. The trained model achieved an accuracy score that exceeded 90%. BotFlowMon (Feng et al., 2019) is a bot detection system based on Multilayer Perceptron (MLP) and CNN algorithms that classify SMB traffic using NetFlow and transaction data. The method classified social bots, including chatbot, post bot, crawler bot, and hybrid bot, with a precision of 98.87%. This research (Kiran et al., 2019) presented a methodology to identify Twitter SMBs based on behavioral features and an ANN model that achieved a best accuracy of 97.79%. The authors of this article (Wu et al., 2021) developed a system using Deep Neural Networks (DNN) and Active learning to identify SMBs in Sina-Weibo. The authors used user information, content, temporal, and interaction features for this aim. Therefore, to build a large-scale dataset, they used active learning to augment the size of the labeled dataset, which achieved a balanced dataset of up to 300,000 samples. Then they designed a model called RGA (ResNet & bidirectional Gated recurrent unit, & Attention), which included ResNet, Bidirectional Gated Recurrent Unit (BiGRU), and an attention layer. The proposed model reached an accuracy score of 98.87%. SATAR (Feng et al., 2021) is a self-supervised technique used for the representation learning of user information, network features, and the semantic information of tweets. Therefore, at the tweet level Word2Vec embedding was used to make the vector representation of each tweet, and the LSTM model was used as a classifier. As a result, SATAR outperformed other cutting-edge systems in detecting social bots, achieving an accuracy of 98.6% and the ability to generalize this approach to all Twitter bots. Bot-DenseNet (Martin-Gutierrez et al., 2021) is a bots detection approach based on RoBERTa embedding layer. The proposed system received an F1-score of 77% based on profile metadata and content features extracted from a Kaggle dataset. In this work (Lingam et al., 2020), the authors studied Twitter botnets and proposed Social Botnet Community Detection (SBCD) system based on graph theory and Deep Autoencoder. For this aim, they compared various similarity-based graphs of users' behaviors, including content, URL, interest, and social interaction similarity. As a result, SBCD achieved a precision of 90%. In (Fazil et al., 2021); a combination of Bi-LSTM and CNN models with an attention layer was proposed for social bot detection. The models were trained with five datasets, and the results showed that the approach outperformed the existing methods with an accuracy of 99.83%. In this work (Beskow and Carley, 2018), the authors studied the automated communication in OSN and proposed a detection system using snowball sampling and RF algorithms. The network features provided relevant information that influenced the model's precision, which reached 92%. The authors of the study (Kumar et al., 2021) used CNN, LSTM algorithms, and BERT for feature extraction to implement a DL-based model to detect SMBs. Based on users' posted content on Twitter, the suggested system achieved an accuracy of 97%. To address the problem of imbalance and the lack of large datasets needed to train the DL algorithm, the authors of (Wu et al., 2020) combined the enhanced Conditional Generative Adversarial Networks (CGANs) with the Gaussian Kernel Density Peak Clustering algorithm (GKDPKA) to prevent the production of noise during data augmentation. They used Wasserstein distance with a gradient penalty to overcome the model collapse and gradient

vanishing of the CGAN model. As a result, CGANs achieved good precision with a score of 97.29% compared to other oversampling techniques such as ADASYN, SMOTE, and Random. Using one-class classifiers does not require examples of outlier behavior to distinguish between bots and human accounts.

#### • Hybrid detection systems

Toward understanding the behavior of malicious SMBs (Bello et al., 2018), the authors suggested a sentiment analysis based on network, content, and sentiment features. Then they analyzed services that create social bots like labnol, botlibre, cheapbots-donquick, jetbots, and botize. Thus, for the experiments, they mined the data using the Twitter API to obtain a maximum of 3200 content posts from every account. Then, they extracted the content features set using bi-gram and Latent Semantic Analysis (LSA). To reduce the number of features to 11 features, linear SVM was used as a feature selection method. As a result, the DT achieved a precision of 98%. In this research, the authors presented an SMBs detection system based on Deep Forest (DF). They used user information, content, and statistical features. The trained model reached an accuracy score of 97.55%. Bot2Vec is a bot detection approach based on network representation learning (Pham et al., 2022). The authors used the skip-gram method to preserve the user's local neighborhood relationships and the intra-node structure of the user node while learning the mapping of OSNs. This technique used only the user's profile without additional features, and as a result, Bot2vec reached a ROC score of 97.68%. Deep-Friend (Wanda and Jie, 2021) is an approach that classifies the malevolent nodes in OSN using Dynamic Deep Learning (DDL) and link features. The system detected malicious nodes with an AUC of 99.70% using the Flickr dataset and 99.56% using a Facebook-based dataset. In this study (Mazza et al., 2019), the authors suggested a system to detect a botnet called Retweet Buster (RTbust) based on temporal patterns extracted from content features. They did this by extracting the LSTM algorithm from the retweets time series and then clustering the group of accounts using the hierarchical DBSCAN clustering algorithm. Therefore, the RTbust approach achieved an F1-score of 87%.

#### • Graph-based detection systems

BotRGCN (Feng et al., 2021) is a detection system based on the Relational Graph Convolutional Network (GRCN) model. The approach used user information, numerical metadata (like the number of followers, the number of followings, the number of active days, etc.), and content features. The experiment was conducted using the TwiBot-20 dataset (Feng et al., 2021), and BotRGCN achieved an accuracy of 84.62%. This work (Wang and Paschalidis, 2017) is a community-based approach that uses both a flow-based approximation that estimates the histogram of quantized flows and a graph-based method that approximates the degree distribution of interaction graphs between nodes, including Erdos-Renyi graphs and scale-free graphs. For the experiment, the authors used the CTU-13 botnet dataset (García et al., 2014), and the proposed system could detect social botnets with a precision of 86%.

### 3.2. Sybil and cyborg detection techniques

Malicious SMBs create profiles on OSNs under different identities to operate malicious activities. The number of accounts they create depends on the purpose of the attackers and the OSN community they target (AlRubaian et al., 2015). Fake or Sybil accounts are both fictive identities.

## • ML-based detection systems

In this paper (Alsaleh et al., 2014), the authors compared the existing Sybil detection systems and explored the Sybil features for an accurate classifier. Data was collected from Twitter accounts using the Twitter REST API and Twitter4j, then manually labeled as human, bot, or hybrid accounts. The proposed MLP model with a gradient descent optimization algorithm achieved an F-measure of 95.1%. *Integro* (Boshmaf, août 2016) is an extension of Syrank that distinguishes between real and fake accounts by ranking the real ones higher than the fakes. *Integro* initially begins by spotting candidate victims through user activity via the RF algorithm, and then it assigns a low weight to edges linked to victims. They used real data with more than 15 million active users to validate the model. The proposed approach achieved higher performance than Syrank, with an AUC score of 76%. In this paper (Yang et al., 2014), the authors studied the characteristics of fake accounts to understand the behaviors of Sybils in Renren. Therefore, they built a dataset of 650,000 Renren Sybils and applied the SVM model to achieve an accuracy of 99%. The study's findings showed that Sybil accounts without explicit connections operate in combination to mount attack campaigns. In this study (Al-Qurishi et al., 2018); the authors proposed a Sybil identification technique based on Feed Forward Neural Network algorithm (FFNN) and using user information, content, and network features. In addition, they used a network-based mechanism that leveraged a connection matrix and a behavior-based mechanism that collected the user activities on the OSN. All these components operated systematically to investigate and evaluate the OSN accounts. The FFNN-based model reached an accuracy of 86%. This research (Al-Qurishi et al., 2018) presented a system based on the RF model to identify malicious behaviors of Sybil accounts based on content features, user information, temporal and network features, and user activities. The system provided three levels of analysis: network level, content level, and user level. Indeed, the authors collected data related to the US elections from YouTube and Twitter. Then, applied random sampling techniques to the training set. The RF-based model achieved an accuracy of 96%.

## • Graph-based detection systems

SybilGuard (Yu et al., 2008) performs lengthy random walks  $O(\sqrt{n} \log n)$  that must intersect with the node to accept a suspect node as honest. But the Sybil admittance is nonetheless constrained by SybilLimit (Yu et al., 2008) by  $O(\log n)$  per attack edge. SybilGuard and SybilLimit's limitations are: related to the number of attack edges, which had a high false negative rate; when the labeled nodes are severely imbalanced, the number of accepted sybils significantly rises. SybilInfer (Danezis and Mittal, 2009) is a threshold based on a probability system that provides strong assurances only on low attack edges and is independent of the topology of the adversary region. It also permits detecting a more significant number of attacker nodes than SybilLimit. In this research work (De Meo et al., 2011), the authors suggested community detection in large networks based on the Louvain method and computing the network's pairwise distance between nodes. Edges are weighted using a global feature that measures their ability to spread information throughout the network. As a result, the proposed system outperformed the state of art approaches. SybilRank (Cao et al., 2012) is an optimum Sybil detection method implemented using Barabasi's scale-free model. It exceeded the previous works by building a graph with fewer weights, requiring a longer mixing time for the Sybil region. And assuming that trustworthy users will not engage with unknown accounts. SybilRank detected Sybils with a TP over the score of 85%. SybilBelief (Gong et al., 2014) is a semi-supervised system implemented using Mar-

kov random fields (MRFs) and loopy belief propagation (LBP) to recognize Sybil nodes. A sample of benign and Sybil nodes was used as input, and it propagated the label from the input data to other nodes in the network. SybilBelief is resilient to noise and can identify Sybil nodes with an accuracy of 100%. SybilSCAR (Wang et al., 2017) is Local rule-based propagation used in OSN for Sybil detection. It combines loopy belief propagation with random walks. To train SybilSCAR, a sizable training dataset was necessarily required. As a result of the experimentation, the SybilSCAR overperformed the SybilRank with an AUC score of 100%. This article (Ma et al., 2014) provided an overview of the current online defense methods in OSN security. It proposed Sybil-Resist, a random-walk-based technique that can identify the Sybil nodes and the Sybil region. The achieved detection rate of 99.2%, which outperformed the state-of-the-art approaches in both detection accuracy and time of execution. SybilFrame (Gao et al., 2018) is a system that detects Sybil attacks by using Pairwise Markov Random Field. The authors validated the proposed approach using a synthetic and real network topology with 20 million nodes and 265 million edges. They showed that SybilFrame could detect Sybils better than structure-based approaches, with an accuracy rate of 95.4%. SybilRada (Mulamba et al., 2016) is a Sybil detection system based on the graph structure of OSN. To train the model, the authors utilized both the weights of edges and similarity values of the nodes on real and synthetic datasets. The finding is that SybilRadar detected Sybils with an AUC rate of 95% even when the attack edges between Sybils and benign are counted in tens of thousands (Cresci et al., 2015). This study (Zegzhda et al., 2017) suggested a system to identify active cyborgs in the OSN VKontakte. By implementing an ANN-based model trained using the user information. The experiment reached an AUC score of 73%. In this paper (Kharaji and Rizi, 2014), the authors proposed a Markov clustering technique to detect cloned profiles based on the network structure of OSN. Similar profiles are gathered into a mutual community, then the relationship strength of all real profiles is rated, and the profiles with fewer values are considered cloned profiles. The finding of this work showed that the proposed system could detect cloned profiles with a TP of 92%. Vote Trust is a graph structure-based technique for Sybils detection. Particular users in a group vote for the confidence of a new user that joined the group and then transmitted their voting to the newly joined user. Here, an interest group refers to a group of smartphones that have one or more common sensing assignments. Trust is then calculated based on the user's acceptance and rejection of their friend requests and their influence. Trust is also computed globally from the votes of the whole network's nodes. In this paper (Caruccio et al., 2018), the authors introduced a mechanism to discriminate real accounts from Sybil accounts on OSN by extracting Relaxed Functional Dependencies (RDF) and big data to characterize the typical patterns of fake accounts. For this aim, they used three datasets: fake accounts, verified accounts, and real accounts.

## • DL-based detection systems

DeepProfile (Wanda and Jie, 2020) is a Dynamic CNN (DCNN) based system that identifies fake accounts using user metadata, content, and temporal features. In addition, they suggested a new pooling layer to enhance the effectiveness of the DCNN model in the training process. The proposed algorithm achieved a precision of 94%. This research work (Benabbou et al., 2022) proposed a system to recognize fake accounts using the BiGRU model and GloVe for word embedding presentation. They trained the model on the BiGRU algorithm based on content features. Therefore, the model outperformed the LSTM and the CNN algorithms and achieved an accuracy of 99.44%.

### • Hybrid detection systems

The classification system presented in (Chu et al., 2010) consisted of the following four elements: (1) an entropy-based component, (2) an ML-based component, (3) a user's information component, and (4) a component to make decisions. Through experimentation, the proposed system could categorize tweets with a TP rate of 82.80%. In this paper (Chu et al., 2012), a probability-based method was developed to classify Twitter accounts as human, bot, or cyborg based on user information, behavior, and content features. For the experiment, they scraped 50,000 accounts from Twitter to implement the system, which was able to classify the Twitter accounts using the Bayes theorem and RF with a TP rate of 96%. In this work (Shetty et al., 2021), the authors proposed a hybrid system to detect fake accounts using SybilGuard, engagement rate, and profile statistics. This approach achieved an accuracy of 95.34%. This work (Khaled et al., 2018) aims to detect Twitter's Sybils based on the SVM-Neural Network (SVM-NN) algorithm and used user information. They applied feature selection techniques and dimension reduction techniques to select effective features. The SVM-NN detected fake accounts with an accuracy of 98.3%.

### • Topic modeling detection systems

HawkesEye (Dutta et al., 2020) detects Sybils based on temporal features, the time series behavior of the retweeting, and the topic modeling of content features using LDA. However, the proposed model outperformed the state-of-the-art approaches with a precision of 96.4%.

### 3.3. Stegobot detection techniques

The stegobots are malicious SMBs that use steganographic images shared by OSN users to build social botnets (Venkatachalam and Anitha, 2017). This paper (Natarajan et al., 2015) proposed a methodology based on the Adaboost algorithm to recognize stegobot hosts in OSN using stegobot communication analysis. The study showed that stegobots have a unique design and implementation. The traffic patterns of stegobots and benign accounts can be detected using a multilevel analysis of the profile information and infected pictures posted on Flickr. The approach achieved an accuracy score of over 97% and an FP rate under 3%. The authors of this work (Venkatachalam and Anitha, 2017) introduced a new host-based solution to detect stegobot accounts and stegobot botnet activities based on the NB algorithm. They used user information and content features, including malicious and steganographic images extracted from Flickr, Google+, and social network features. Therefore, experimental results showed that combining content features with network features increased the results of the NB algorithm and achieved an accuracy of 96.24%.

### 3.4. Political bots detection techniques

A political bot is a software used and controlled by governments and political parties to manipulate public opinion and amplify polarization and sentiment by using fake likes, posting content, tweeting, and retweeting (Hagen et al., 2022); (Fernquist et al., 2018). Their activity revolves around the most important political events and spreads misinformation, viral content, and fake news on OSN using memes and deep fake videos (Zannettou et al., 2019). In this research study (Bruno et al., 2022), the authors analyzed social bots' behavior during the United Kingdom (UK) elections of 2019 by analyzing online talks and tweets and then

examining both the temporal pattern and statistics of the content posted about Brexit. The finding is that social bots participated in the social debate on Twitter, especially on the days near the UK elections, in which the discussion of bots increased; it jumped from [100000, 300000] in regular days to 900,000 during the election and then returned to the normal values observed before the elections. The authors of this paper (Hurtado et al., 2019) analyzed the political discussion on Reddit by building the network structures of this OSN. Unlike other OSNs, such as Facebook and Twitter, Reddit users register for a topic and then participate in the discussion by posting comments. This analysis showed that a group of fully connected members whose names contain the word "bot" have bot-like behavior. In this work (Hagen et al., 2022), the authors examined the influence of SMBs on the political discussion in OSNs. They used content and sentiment features to analyze the network structure of the active accounts in the political debate. As a result of the study, SMB influenced political debate using fake likes and network sentiment amplification.

### • ML-based detection systems

In this study (Castillo, et al., 2017), the authors analyzed the activity performed by electoral candidates in the Chilean presidential campaign during the 2017 election for political bot detection. To do so, they extracted from Twitter user metadata, metadata of friends' accounts, network, temporal, tweet, and sentiment features to train the SVM model. SVM achieved over 80% for the training set and decreased to less than 60% for the test set. This work proposed (Beğenilmiş and Uskudari, 2018) a system based on the RF model to detect the coordinated fake news activities promoted by human and social bots during the United States (US) presidential election of 2016. For the experiments, they collected data based on trending hashtags, and a set of data included user information and temporal features. RF achieved an average accuracy of 95%. To detect political bots, this work (Pratama and Rakhmawati, 2019) used an RF-based model trained using the comments posted by the supporters of Indonesian presidential candidates on Twitter. The data were extracted from Twitter using filters such as Joko Widodo (@jokowi) and Prabowo Subianto (@prabowo). The RF reached an accuracy score of 74%. BotCamp (Abu-El-Rub and Mueen, 2019) is a framework based on the Boosted DT algorithm for political bot detection based on temporal, content, and network features. They automatically collected data from the trending hashtags and applied a graph clustering of retweets, mentions, hashtags, and temporal characteristics. BotCamp detected political bots with an accuracy of 87.5% and a precision of 98%. This study revealed that the studied political SMBs were similar in temporal correlation, sentiment, and topic correspondence. This work (Sayyadiharikandeh et al., 2020) suggested ensembles of specialized classifiers (ESC) to identify political bots using user information, behavior, friends, networks, temporal, and content features. The proposed system is an update of the Botometer, which achieved an AUC rate of 99%. In this study (Teljstedt et al., 2015), the authors suggested a political bot detection approach that copes with the complexity of manual annotation of massive datasets, which requires additional effort and resources. To achieve this goal, the authors developed a semi-automatic annotation of datasets based on the RF model. For the experiment, the author used a Russia-Ukraine conflict dataset extracted from Twitter and proposed new features like "inter-tweet-content similarity" (ITCS), "Pearson's Chi-Square Statistics of Uniformity" (PCU), "Rao's Spacing Statistics of Uniformity" (RSU), "inter-tweet-content regularity" (ITC-R), "inter-tweet-delay shape" (ITD-S) and "inter-tweet-delay regularity" (ITD-R). The RF model achieved a precision of 97.4%.



### 3.5. Spam bots detection techniques

Spam bots are malicious SMBs that disseminate malicious Uniform Resource Locator (URL), spam messages, and comments, including misinformation and fake news (Ellaky et al., 2021). Spam bots routinely abuse OSN users and perform a variety of online crimes, including trolling, spreading rumors, stalking, and cyberbullying.

#### • ML-based detection systems

In this study (Miller et al., 2014), the authors dealt with detecting spam bots as an anomaly problem. They extracted 95 one-gram attributes of tweets and user information to do so. Therefore, to optimally handle the continuum of tweets, two kinds of stream clustering techniques were applied, StreamKM++ and DenStream, which were both modified to improve spam recognition. The two algorithms clustered benign users and considered the outliers as spam bots. Using these algorithms together gave a recall of 100% and a false positive of 2.2%. This research (Inuwa-Dutse et al., 2018) proposed a system based on Gradient Boosting (GB) model for spam bot detection that aimed to understand spam behavior. To collect data, they applied LSA on a honeypot dataset to extract concepts related to spam bots' posting. The Recursive Feature Elimination (RFE) method selected the most significant characteristics, excluding historical content. GB algorithm performed the best accuracy of 98.97%. In this research work (Andriotis and Takasu, 2018), the authors used AdaBoost to classify Twitter spam bots based on content features, user information, and sentiment features. They used four known datasets ICWSM17 (Varol et al., 2017), RAID11 (Liu et al., 2017), WWW17 (Yang et al., 2013), and Cresci-17 (Cresci et al., 2017). As a result, AdaBoost achieved a precision of 95%. The Social Spammers and Spam Messages in Microblogging Detection (S3MCD) (Wu et al., 2016) is a model based on the Alternating Direction Method of Multipliers (ADMM) algorithm. The approach used content and user information features and three relation matrices: user-message links, user-user links, and message-message links. The system exploited the posting links between the OSN accounts and the content posted to combine SMBs detection with spam content detection. The Term Frequency Inverse Document Frequency (TF-IDF) vector was employed to extract the feature vector. Logistic regression (LR) was used to assign scores to microblogging accounts and the posted content. As a result, S3MCD reached a precision score of 92.34% in spam bots detection. The authors of (Liu et al., 2017) studied the imbalanced spam and legitimate accounts. Therefore dataset, a fuzzy oversampling technique (FOS) was used to fix the problem of unbalanced data. The experiment results showed that this system improved the detection of spam bots using imbalanced datasets and an RF-based model, which achieved a TP score of 78%. Authors in (Loyola-Gonzalez et al., 2019) proposed a system based on a contrast pattern model to detect spam bots. The suggested framework conducts the classification in four steps: 1) data extraction from Twitter that includes user information, content, timing, and sentiment features (sub, irony, agreement, confidence, feeling) extracted using the Meaning Cloud API; 2) Multiple different DT generation from the proposed feature representation using different quality measures to evaluate the splits, where the contrast patterns are extracted from each DT; 3) identification of the best contrast patterns in features in the training set; 4) accounts classification. The RF performed higher than the reviewed techniques, with an AUC of over 99.56%. The authors in (Murugan and Devi, 2019) proposed a system for Twitter spammers detection based on RF algorithms. To extract the features, they used a hybrid technique by employing both LR with Principal Component Analysis (PCA) to reduce the dimension of the feature set. The overall

performance of the selected features from the user profile showed an increase in detection rate, reaching an F-1 measure score of 92%.

#### • DNA-inspired detection systems

To deal with the fact that bots can evade detection by randomizing their actions, the authors of this work (Cresci et al., 2016) proposed a DNA-inspired model to detect spam bots. This approach encoded the behavior of OSN users in a sequence of characters that represented the behavioral lifetime. Based on the Longest Common Substring (LCS) and a similarity measure applied to each digital DNA sequence to cluster the group of spam bots. They considered that a cluster of users that have a similar series of actions are spam bots.

#### • Hybrid detection systems

This study (Fazil and Abulaish, 2018) proposed a hybrid approach to detect spam bots on Twitter based on tweets, network attributes, and user information. In addition, they introduced a new feature to conduct hybrid detection: community and interaction features. For the experiment, they used the social spammer dataset proposed in (Cresci et al., 2016) to train the RF algorithm, which exceeded the baseline models with an accuracy of 97.9%. The Spammer Detection Hybrid Model (SDHM) (Liu et al., 2014) is a hybrid model for social spammers' detection in Weibo based on topic modeling. They used LDA for topic extraction from content features, and content similarity was computed using the Longest Common Subsequence (LCS). In addition, user information and network features were investigated. The authors support that SDHM can detect spam bots with only one user's posting behavior and an F-measure of 91.8%. The experiment results showed that the proposed system achieved an accuracy score of 92.9% without requiring historical data or user information. In this paper (El-Mawass et al., 2018), the authors employed the Markov random field (MRF) technique to fix the evasion-induced errors in spam bot detection. They used the graph model in conjunction with the SVM model, which outperformed traditional classifiers with a precision of 95.2%.

#### • DL-based detection system

The results revealed that 79% of evolved spam bots evade state-of-the-art detection systems like in (Cresci et al., 2017) and (Miller et al., 2014). In this work (Lingam et al., 2021); the authors used Deep Q-learning (DQL) and Particle Swarm Optimization (PSO) to detect spam bots on Twitter. The features used are user information, content, and temporal features. Moreover, the study proposed an algorithm called Spam-Influential User Identification Integrated into Influential Community Detection (SIU-ICD). The SIU-ICD reduced the dissemination of spam influential communities on OSNs and achieved a precision of 94%. The authors in (Ali Alhosseini et al., 2019) presented a spam bots detection system using inductive representation learning and user information. The Graph Convolutional Neural Networks algorithm (GCNN) exhibited the best performance compared to MLP and Belief Propagation with a precision of 89%. This paper (Wei and Nguyen, 2019) aims to detect Twitter spam bots based on Bi-LSTM and content features. They used Word Embedding for feature extraction from users' tweets. This work (Mateen et al., 2017) presented an approach for spam bots detection based on the J48 algorithm to detect Twitter spam bots using user metadata, content, and network features. The J48 algorithm was able to detect spam bots with a precision of 97.6%. The spam bot detection approach proposed in (Zhao et al., 2020) used a heterogeneous stacking-based ensemble learning method. To adjust the weights of the classifiers' prediction



results, they used cost-sensitive learning in the training stage of the DNN-based model that achieved a precision of 70%.

#### • Graph-based detection systems

This research work (Lingam et al., 2018) presented a social spam-botnet detection technique based on content and network features and the trust concept. The trust value between OSN users is calculated based on Bayesian theory, and the value of indirect trust among neighboring users is calculated based on the Dempster-Shafer theory. The incorporation of both parameters improved the accuracy of the detection by a score of 85%. Based on Triad Significance Profile (TSP), Social Status, and Cascaded filtering, this study (Jeong et al., 2016) proposed spam bot detection mechanisms. These approaches analyzed and exploited social network properties. This method's main advantage is scalability. Instead of analyzing the entire network, it examines user-oriented “two-hop” social networks. This approach detected spam bots with a TP score of 96.3%.

#### • Topic modeling based detection systems

In this paper (Koggalahewa et al., 2022), the authors implemented an unsupervised technique to detect spam bots using LDA and similarity-based clustering. They calculated the matching acceptance between OSN users based on frequent interests in specific topics between two users. Therefore, they used LDA for topic modeling and then trained the model, which achieved an accuracy of 96.9%.

#### • Genetic-based detection systems

GenBot (Cresci et al., 2019) is a genetically-based system that generated synthetic spam bots by generating digital DNA sequences of user information and content features by encoding user posts into chronological segments of action using a digital DNA technique. Identifying behavioral evolutions and mutations of spam bots in OSNs was made possible by converting user information and content features to digital DNA sequences. Transforming user information and content features to digital DNA sequences permitted the identification of behavioral evolutions and mutation of spam bots in OSNs. In (Cresci et al., 2019), authors applied an ad-hoc genetic algorithm to detect social spammers based on digital DNA. To represent the digital DNA of an OSN user, they embedded the behavior of each OSN account as a numerical DNA sequence. The behavior of clusters of spam bots was modeled via their numeric DNA and compared to a set of benign accounts. The finding, it is possible to examine the ability of spam bots to escape the existing behavior-based detection techniques by switching parameters of the genetic model. In this work (Akiyama et al., 2017), the authors studied the URLs included in spam content that redirects users to malicious websites based on a URL tracking system using Domain Genetic Algorithms (DGA) to blocklist the malicious links. This

approach is a honeypot-based system that prevents URL redirection with an accuracy of 96.50%.

#### 3.6. Game bots detection techniques

A game bot is an SMB that can play OSN games instead of human users to gain more money and prizes by playing continuously and winning more items that human users cannot. This study (Kang et al., 2016) aims to propose a system based on the RF model to detect social game bots active in Massive Multiplayer Online Role Playing Games (MMORPG). First, the authors investigated the game bots' behaviors and discovered they were engaged in repetitive trading for real-time money. They used different user data from MMORPG enterprises and focused on behavior-based data, including activity logs, chat contents, user information, and network features. The approach achieved an accuracy of 96.06%.

In this chapter, we have reviewed the SMB detection techniques proposed by researchers, including all existing types of SMBs, such as social bots, Sybils, cyborgs, stegobots, political bots, spam bots, and game bots. Based on these state-of-the-art works, we will address the research questions of this SLR and outline open questions for future work.

### 4. Review analysis

This section discusses the key findings of our systematic literature review, which were derived from the most important studies published between 2008 and 2022. One hundred forty-four articles were regarded as primary because they included several experiments in SMB detection. Therefore, it investigates the studied approaches by analyzing the techniques used by the researchers to build SMB detection systems from five perspectives: (1) The type of bots targeted by the SMB detection techniques; (2) The in-depth study of the algorithms and the technique used for the SMBs detection, performance comparison, discriminative features, and the dataset that gave better results for SMBs detection. (3) The classification of the features used by the SMBs detection techniques; (4) The influence of feature engineering techniques on the accuracy of the models; and (5) The analysis of the data sources used for SMB detection, including the type of OSNs, the presentation of dataset details, and dataset issues such as dataset size and labeling problems. The 5th research question raised at the outset of this study will be addressed by conducting a quantitative and qualitative analysis of the overall experiment results. This research will go over each stage of the lifecycle of an SMB detection model and provide insights for new research studies to improve the detection of SMBs.

#### 4.1. Profiles and social media bot types (RQ1)

According to the literature, there are different types of profiles on the OSN; the legitimate profile is the normal profile of accounts belonging to real people conducting legitimate activities. This category of profiles also includes the verified profile, which is an

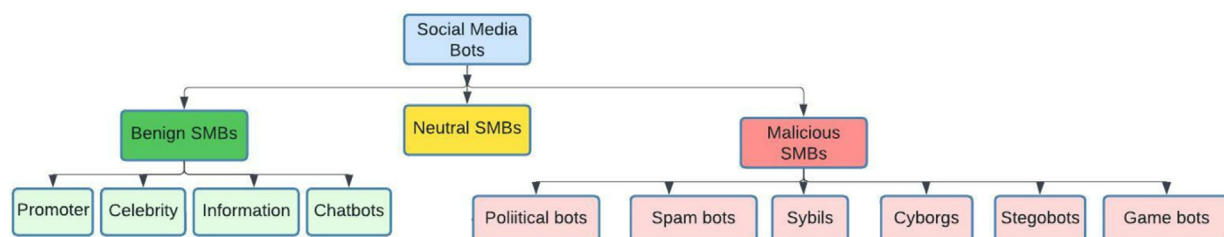


Fig. 3. Taxonomy of social media bots.

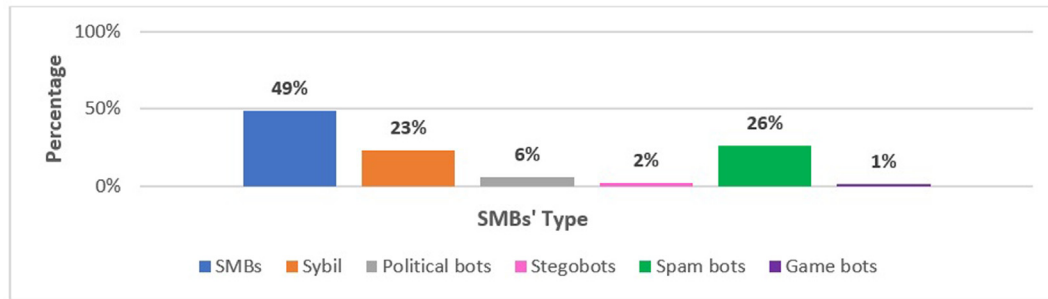


Fig. 4. Type of bots studied in the state-of-the-art works.

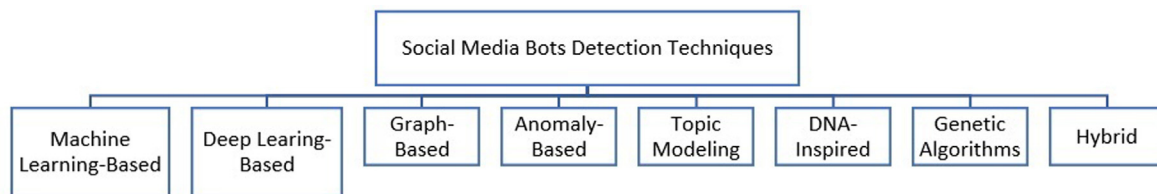


Fig. 5. Taxonomy of SMBs detection approaches.

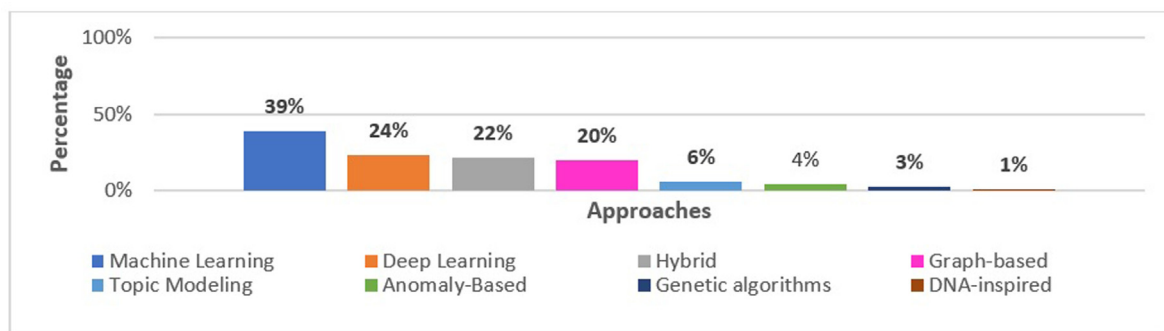


Fig. 6. SMBs Detection Techniques.

account belonging to companies, celebrities, or public figures whose profile authenticity has been confirmed by OSN. The cloned profile is the account whose identity and metadata have been stolen from a legitimate profile to conduct malicious activities. Fake accounts are pseudo-identities used to infiltrate the OSNs; they can be managed manually by a person or automatically by a bot master. Sock puppets are manually controlled OSN profiles, and they are designated to support something or someone on the OSN. Therefore, troll profiles are the politically engaged Sock Puppets controlled by government proxies or political parties to spread misinformation, viral content, and fake news to influence OSN users during political events and elections.

An SMB can mimic human behavior on OSNs and perform automated activities, including posting textual content, photos and interacting with other users. In general, an SMB supports two kinds of generic operations in OSN. The first one is social interactions, which are the operations of writing and reacting to content posted by other OSN users. The second is social networking operations that are used to alter the social structure of a group of users by adding or following other accounts. Therefore, each social bot requires a profile in an OSN to launch these activities. An SMB can be classified according to its intention and behaviors, whether benign, neutral, or malicious (Stieglitz et al., 2017). Benign bots automatically post alert tweets, disseminate news (such as news bots), interact with users, and respond to

their needs (such as chatbots). Neutral bots post or repost jokes, such as “nonsense bots” that share nonsense content. Malicious SMBs are known as “zombies” or “slaves”, which a bot master controls to perform automatic operations and coordinate attacks. The most known malicious SMBs are spam bots, political bots, Sybils, cyborgs, stegobots, and game bots. Based on the literature review, we classified malicious SMBs into six types according to their severity level, as shown in Fig. 3. In this taxonomy, we defined four categories for benign SMBs and six main categories for malicious SMBs.

Despite the variety of bots operating in different OSNs, many works have been devoted to SMB detection and classification. Most of these works are based on a binary supervised learning algorithm that classifies the OSN accounts as (human or non-human) or (Human legitimate or malicious bots). These works proposed generic SMB detection without specifying the type of social bot. Fig. 4 depicts the percentage of papers dedicated to each category of SMBs.

From Fig. 4, we observe that generic SMB detection is the most common approach carried out in the literature, followed by spam bots, Sybils, and political bots detection approaches. Few studies were conducted to detect stegobots and game bots. This can be explained by the fact that the researchers were initially more concerned with figuring out whether a particular account was legitimate or not, as well as whether it was a real account or a bot account. The growth of this field of study has made it possible to

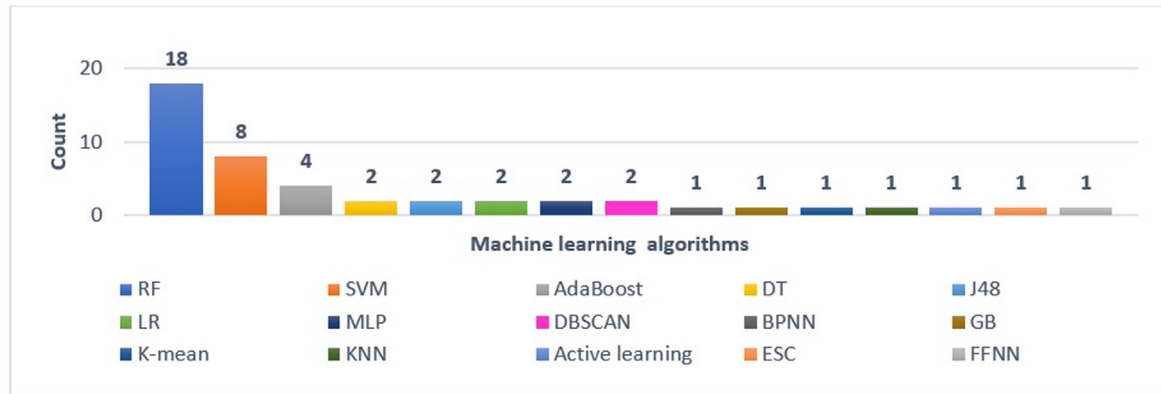


Fig. 7. ML Algorithms used for SMBs detection.

**Table 3**  
ML-based detection systems.

| Algorithm                        | References                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Max Performance                               | Dataset of Max Performance                     | Features of Max Performance                                           |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------|
| AdaBoost                         | (Dickerson et al., 2014) (Morstatter et al., 2016) (Natarajan et al., 2015) (Andriotis and Takasu, 2018)                                                                                                                                                                                                                                                                                                                                                                                | Accuracy: 93.89%,<br>F1-score: 95.06%         | ICWSM17, RAID11, WWW17, Cresci-17              | User information, sentiment, and content features                     |
| SVM                              | (Ikram, et al., 2017) (Dorri et al., 2018) (Beskow and Carley, 2019) (Fernandes et al., 2015) (Bello et al., 2018) (Yang et al., 2014) (Castillo, et al., 2017) (El-Mawass et al., 2018)                                                                                                                                                                                                                                                                                                | AUC: 100%<br>Recall: 99.8%                    | Scraped Dataset from RenRen                    | User information, content, network, and temporal features             |
| DT                               | (Bello et al., 2018) (Ellaky et al., 2021)                                                                                                                                                                                                                                                                                                                                                                                                                                              | Precision: 100%                               | Cresci-17                                      | Content features                                                      |
| RF                               | (Davis et al., 2016) (Jr et al., 2018) (Ellaky et al., 2021) (Beğenilmiş and Uskudarli, 2018) (Ji et al., 2016) (Alothali et al., 2021) (Igawa, et al., 2016) (Rodríguez-Ruiz et al., 2020) (Pratama and Rakhmawati, 2019) (Beskow and Carley, 2018) (Boshmaf, août 2016) (Al-Qurishi et al., 2018) (Teljstedt et al., 2015) (Chu et al., 2012) (Loyola-Gonzalez et al., 2019) (Fazil and Abulaish, 2018) (Kang et al., 2016) (Murugan and Devi, , 2019) (Jin Dan and Teng Jieqi, 2017) | Precision: 100%                               | Cresci-17                                      | Content features                                                      |
| BPNN                             | (Mazza et al., 2019)                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Accuracy: 98.6%                               | –                                              | Network features                                                      |
| J48                              | (De Meo et al., 2011) (Yu et al., 2008)                                                                                                                                                                                                                                                                                                                                                                                                                                                 | FP: 1.7%<br>F-measure: 97.9%                  |                                                | User information, content, and network features                       |
| GB                               | (Inuwa-Dutse et al., 2018)                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Accuracy: 98.97%                              | Honeypot dataset                               | User information, content features                                    |
| GBT ensemble learning algorithms | (Zannettou et al., 2019)                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Recall: 85%<br>Accuracy: 86%<br>F1-score: 83% | Scraped dataset from Twitter                   | User information, content, and temporal features                      |
| KNN                              | (Mohammad et al., 2019)                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Recall: 96%                                   | Scraped dataset from Twitter                   | User information, content, and temporal features                      |
| MLP                              | (Ali Alhosseini et al., 2019) (Alsaleh et al., 2014)                                                                                                                                                                                                                                                                                                                                                                                                                                    | Precision: 98.87%                             | NetFlow and transaction dataset                | User information, content features                                    |
| ESC                              | (Sayyadiharikandeh et al., 2020)                                                                                                                                                                                                                                                                                                                                                                                                                                                        | AUC: 99%                                      | Pron-bots, Cresci-17                           | User information, content features                                    |
| FFNN                             | (Al-Qurishi et al., 2018)                                                                                                                                                                                                                                                                                                                                                                                                                                                               | TP: 67.7%                                     | Fake followers<br>Scraped dataset from Twitter | User information, content, temporal, linguistic, and network features |

analyze social bots in-depth and comprehend their behaviors related to each type of social media attack and illegal behavior that SMBs engage in.

#### 4.2. Detection approaches and performance (RQ2)

##### 4.2.1. Artificial intelligence methods

This section discusses the different SMB detection approaches and the dataset features used to train models. Researchers have used many methods to build a system to detect malicious SMBs

accurately and prevent their spread. These approaches are ML-based, DL-based, graph-based anomaly-based, topic modeling based, DNA-inspired, genetic-based, and hybrid. To better understand the method academics have suggested for detecting SMBs, we categorize the current SMBs detection methodologies. The taxonomy of these approaches is shown in Fig. 5.

Fig. 5. ML-based methods applied traditional ML algorithms such as SVM, RF, DT, KNN, LR, etc. Therefore, the DL-based methods utilized Deep learning algorithms like CNN, LSTM, MLP, etc. The graph-based approaches are implemented using graph theory

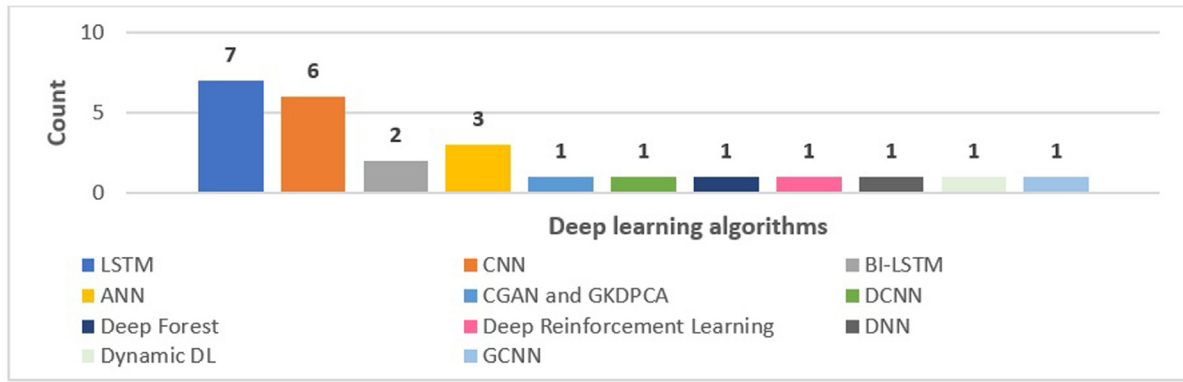


Fig. 8. Deep learning algorithms used for SMBs detection.

**Table 4**  
DL-based detection system.

| Algorithm                     | References                                                                                                                                                   | Max Performance                                         | Dataset of Max Performance                          | Features of Max Performance                               |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------|
| LSTM                          | (Mazza et al., 2019) (Kudugunta and Ferrara, 2018) (Beğenilmiş and 2018) (Feng et al., 2021) (Cai et al., 2017) (Kumar et al., 2021) (Benabbou et al., 2022) | Recall: 100%<br>Accuracy: 100%<br>Precision: 99.8%      | Crisci-17                                           | User information, content features                        |
| BI-LSTM                       | (Beğenilmiş and Uskudarli, 2018) (Loyola-Gonzalez et al., 2019)                                                                                              | Precision: 94%<br>Recall: 97.6%<br>Accuracy: 96.1%      | Cresci-17                                           | Content features, deep linguistic features                |
| CNN                           | (Cai et al., 2017) (Mohammad et al., 2019) (Feng et al., 2019) (Fazil et al., 2021) (Kumar et al., 2021) (Benabbou et al., 2022)                             | Recall: 100%<br>Accuracy: 100%<br>Precision: 99.8%      | Crisci-17                                           | User information, content features                        |
| DCNN                          | (Wanda and Jie, 2020)                                                                                                                                        | AUC: 95.47%<br>Precision: 94%<br>Recall: 93.21%         | Scraped dataset from Twitter                        | User information, content, and temporal features          |
| GCNN                          | (Ali Alhosseini et al., 2019)                                                                                                                                | AUC: 94%<br>Recall: 80%<br>Precision: 89%               | Content polluters dataset Yang et al. 2013          | User information                                          |
| DNN                           | (Cresci et al., 2016)                                                                                                                                        | Precision: 70%                                          | The dataset used in (Chen et al., 2015) Cresci-2015 | User information, content features                        |
| ANN                           | (Mohammad et al., 2019) (Kiran et al., 2019) (Zegzhda et al., déc. 2017)                                                                                     | Accuracy: 97.79%                                        | Scraped dataset from Twitter                        | User information, content features                        |
| Deep Forest                   | (Daouadi et al., 2019)                                                                                                                                       | Accuracy: 97.55%                                        | –                                                   | User information, content features                        |
| CGAN and GKDPCA               | (Kantepe and Ganiz, 2017)                                                                                                                                    | Precision: 97.29%<br>Recall: 97.93%<br>Accuracy: 96.55% | Scraped dataset from Twitter                        | User information, content, and network features           |
| RGA: ResNet, BiGRU, Attention | (Wu et al., 2021)                                                                                                                                            | Precision: 99.33%                                       | Scraped dataset from Sina Weibo                     | User information, content, network, and temporal features |
| Dynamic DL                    | (Wanda and Jie, 2021)                                                                                                                                        | AUC: 99.70%<br>Recall: 90%                              | Flickr, Facebook                                    | User information                                          |
| DRL                           | (Akiyama et al., 2017)                                                                                                                                       | Precision: 94%                                          | Social Honeypot, Cresci-2015                        | User information, content, and temporal features          |
| BiGRU, LSTM, and CNN          | (Benabbou et al., 2022)                                                                                                                                      | Accuracy: 99.44%<br>Precision: 99.25%                   | Cresci-17                                           | Content features                                          |

algorithms like The Random Walk, Loopy Belief propagation, Pair-wise Markov Random Field, etc. The aim is to build the network structure of a group of accounts to detect the Sybil nodes and to prevent the Sybils attacks. The anomaly-based approaches are used to identify OSNs intrusions and attacks conducted by SMBs. This approach is often used during the first stage of SMB detection by clustering the group of accounts with abnormal behaviors. The topic modeling techniques allow the detection of emerging topics in the microblogs of OSNs. These techniques, like LCS, LDA, etc., enable the detection of propaganda, manipulation, and disinformation campaigns carried out by malicious SMBs. The approaches based on DNA-inspired and Genetic algorithms used

the notion of a sequence of interaction and timeline posting and action to detect abnormal behavior of SMBs. So, the hybrid approaches combine several algorithms to operate in a complementary and mutual way to increase the efficiency of malicious SMBs recognition.

Fig. 6 provides a statistical overview of the works based on the different categories of approaches. Therefore, it is noticeable that 39% of these detection systems are based on ML algorithms, then DL algorithms with 24%, and hybrid methods that combine different algorithms with 21%.

The supervised technique based on ML and DL algorithms can overcome the issue of scalability and the high cost by reducing



**Table 5**  
Graph-based detection systems.

| Algorithm                                  | References                   | Max Performance                      | Dataset of Max Performance                                                                                                                                                                                                                   | Features of Max Performance                        |
|--------------------------------------------|------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| SybilGuard                                 | (Yu et al., 2008)            | –                                    | 1 M node from 45,000 Web documents                                                                                                                                                                                                           | Network features                                   |
| SybilLimit                                 | (Yu et al., 2008)            | –                                    | Data crawled from: <a href="https://www.friendster.com">https://www.friendster.com</a><br><a href="https://www.livejournal.com">https://www.livejournal.com</a><br><a href="https://www.dblp.uni-trier.de">https://www.dblp.uni-trier.de</a> | Network features                                   |
| SybilInfer                                 | (Danezis and Mittal, 2009)   | FP < 5%                              | LiveJournal and power law network dataset: 300,000 nodes                                                                                                                                                                                     | Network features                                   |
| SybilRank:                                 | (Cao et al., 2012)           | TP > 85%                             | Facebook, CA-GrQc, CA-HepTh, CA-AstroPh, CA-CondMat, wiki-Vote, soc-Epinions, soc-Slashdot, and email-Enron datasets                                                                                                                         | User information, network, and behavioral features |
| Louvain method                             | (De Meo et al., 2011)        | Normalized mutual information: 81.6% | Facebook dataset                                                                                                                                                                                                                             | Network features                                   |
| SybilSCAR                                  | (Wang et al., 2017)          | AUC: 100%                            | CA-GrQc, CA-HepTh, CA-AstroPh, and CA-CondMat datasets                                                                                                                                                                                       | Network features                                   |
| SybilBelief                                | (Gong et al., 2014)          | AUC: 99%                             | Facebook dataset                                                                                                                                                                                                                             | Network features, User information                 |
| Sybil-Resist                               | (Ma et al., 2014)            | AUC: 82%                             | Small Twitter dataset                                                                                                                                                                                                                        | Network features, User information                 |
| SybilFrame                                 | (Gao et al., 2018)           | FP < 0.3%                            | Facebook dataset                                                                                                                                                                                                                             | Network features                                   |
| SybilRadar                                 | (Mulamba et al., 2016)       | Accuracy: 100%                       | Slashdot dataset                                                                                                                                                                                                                             | Network features                                   |
| Íntegro                                    | (Boshmaf, 2016)              | AUC < 95%                            | Email dataset                                                                                                                                                                                                                                | Network features                                   |
| Similarities-based and relationships-based | (Kharaji and Rizi, 2014)     | Detection rate: 99.2%                | Facebook dataset                                                                                                                                                                                                                             | Network features                                   |
| Trust vote                                 | (Yang et al., 2016)          | Accuracy: 95.4%                      | Massive Twitter dataset                                                                                                                                                                                                                      | Network features                                   |
| Erdos-Renyi graphs and scale-free graphs   | (Wang and Paschalidis, 2017) | Recall ≥ 80%                         | The Cresci-2015 dataset<br>The Elezioni2013 dataset<br>TWT dataset<br>The INT dataset                                                                                                                                                        | Network features                                   |
| Sybil Guard, ML, Loop Belief Propagation   | (Shetty et al., 2021)        | AUC: 76%                             | Pro-Facebook, Pro-Tuenti, Gra-facebookts<br>Gra-facebookrd, Gra-hepTh, Gra-astroph Graph                                                                                                                                                     | User information<br>Network features               |
| Node2Vec and intra-community random walk   | (Pham et al., 2022)          | –                                    | Facebook interactions dataset                                                                                                                                                                                                                | User information<br>Network features               |
|                                            |                              | Precision: 86%                       | RenRen dataset                                                                                                                                                                                                                               | User information<br>Network features               |
|                                            |                              | Accuracy: 95.34%                     | Peking University (PKU) network: friend invitation graph for PKU network 5.01 M complete friend request records, 230 K PKU users                                                                                                             | User information<br>Network features               |
|                                            |                              | ROC: 97.68%                          | CTU-13 botnet dataset                                                                                                                                                                                                                        | Network features                                   |
|                                            |                              |                                      | Scraped dataset from Twitter                                                                                                                                                                                                                 | User information<br>Network features               |
|                                            |                              |                                      | Kaggle and Machine Learning Detecting-Twitter-Bots dataset                                                                                                                                                                                   | Statistical features<br>User information           |

human intervention during the detection process. From the Fig. 6, we observe that 39% of the proposed approaches were implemented using ML algorithms, 24% using DL algorithms, 22% using hybrid, 20% using graph-based algorithms, 6% using topic modeling techniques, 4% using anomaly-based techniques, and then the genetic-based and the DNA-inspired algorithms. The count distribution of the employment of ML techniques in detecting SMBs is shown in Fig. 7.

RF, SVM, and AdaBoost are the three most well-known ML algorithms for social bot detection, as illustrated in Fig. 7. Also, we observe that supervised ML methods are applied to SMB identification more frequently than unsupervised methods. Table 3 presents the ML algorithm, references, greatest performance recorded, training dataset, and features used to obtain maximum performance. The aim of Table 3, evaluate the ML-based techniques used in SMB identification.

The RF algorithm is the most widely used model classifier, as shown in Table 3, and it successfully identified spammers, social media bots, and political bots. For the best accuracy, the RF algorithm reached 100% and a recall of 100% with models trained using the Cresci-17 dataset and content features (Ellaky et al., 2021); (Igawa, et al., 2016). In the second range, the SVM attained an AUC of 100% (Ikram, et al., 2017) and a recall of 99.8% with a model

trained on a dataset containing user information, content, network, and temporal variables collected from RenRen. The performance findings of the additional ML methods, mainly when applied to the Cresci-17 dataset, are encouraging. Supervised ML algorithms have effectively identified SMBs, spam bots, Sybils, cyborgs, and political bots. They only need a few features to identify SMBs. From the state-of-the-art studies examined in the preceding sections and comparing the results presented in Table 3, we discovered that user information and content features are the most frequently used attributes for ML-based detection systems. To identify botnets and steganobots, which require more intricate information like steganographic images and videos that cannot be preprocessed appropriately or trained using ML algorithms. SMBs keep changing their attack strategies to evade detectors. Moreover, the complex experiments involve more significant amounts of data, for example, to reconstruct the structure of the social network or to extract the more advanced characteristics, which increase the complexity of the ML-based model and the complexity of the learning algorithm. Things are very different when it comes to DL-based models. Unlike traditional machine learning, deep learning does not require any feature engineering phase. It can learn the representation from the input data with no preconceived structure. Fig. 8 illustrates the count distribution of the employment of DL techniques to detect SMBs.

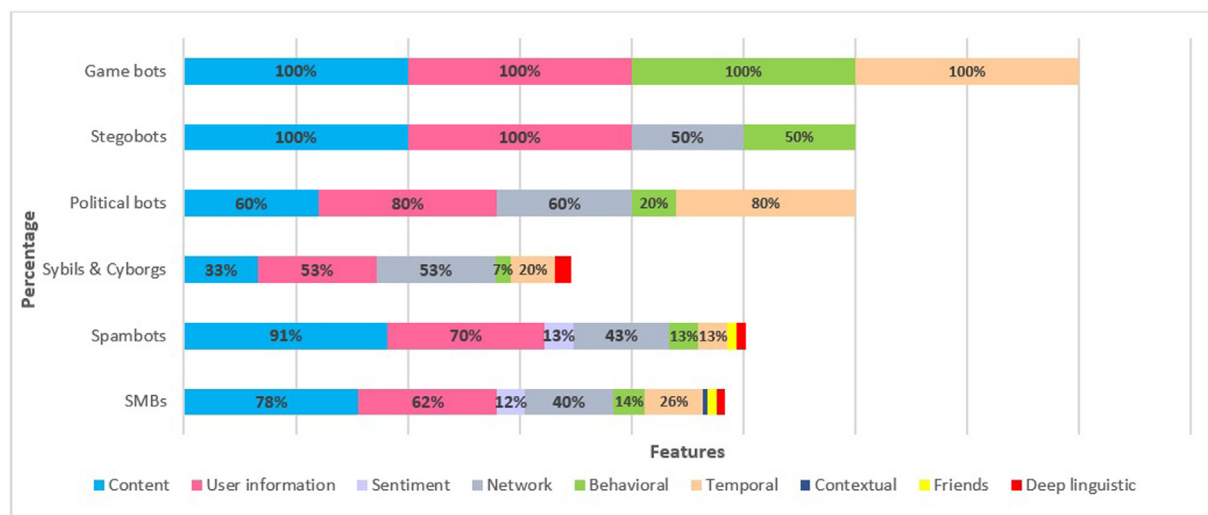
**Table 6**

Other detection systems.

| Approach                                                    | Type of Approach   | References                     | Max Performance                                                           | Dataset of Max Performance                                                                                                                                                 | Features of Max Performance                                                    |
|-------------------------------------------------------------|--------------------|--------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| ActM                                                        | Anomaly-based      | (Davis et al., 2016)           | Recall: 92%                                                               | Scraped datasets from Twitter, Reddit, Stack-Overflow                                                                                                                      | Temporal and content features                                                  |
| Bayes theorem and RF                                        | Hybrid             | (Chu et al., 2012)             | Precision: 93%<br>TP: 96%                                                 | Scraped dataset from Twitter                                                                                                                                               | User information, behavioral and content features                              |
| BotWalk                                                     | Anomaly-based      | (Minnich et al., 2017)         | Precision: 90%                                                            | Scraped dataset from Twitter                                                                                                                                               | User information, content features, temporal, network, and behavioral features |
| Causality-based Clustering, label propagation algorithm     | Hybrid             | (Shaabani et al., 2018)        | Precision: 75%                                                            | ISIS dataset from Twitter                                                                                                                                                  | User information, content, and sentiment features                              |
| Debot and Graph-based clustering                            | Hybrid             | (Abu-El-Rub and Mueen, 2019)   | Recall: 83.6%<br>Accuracy: 87.5%<br>Precision: 98%                        | U.S. election dataset 2016 from Twitter                                                                                                                                    | Content, temporal, network, and sentiment features                             |
| One-class RF                                                | Anomaly-based      | (Rodríguez-Ruiz et al., 2020)  | AUC: 89%                                                                  | Cresci-17 dataset                                                                                                                                                          | Behavioral features                                                            |
| Trust Model & Dempster-Shafer theory                        | Hybrid             | (Lingam et al., 2018)          | Accuracy: 85%                                                             | Cresci-2015 dataset                                                                                                                                                        | User information, content, and network features                                |
| DGA                                                         | Genetic algorithms | (Akiyama et al., 2017)         | Accuracy: 96.50%                                                          | Dataset scraped from Twitter of content injected with URL redirection                                                                                                      | Content features                                                               |
| flow-based approximation & Graph modeling                   | Hybrid             | (Kiran et al., 2019)           | Precision: 86%                                                            | CTU-13                                                                                                                                                                     | Botnet traffic                                                                 |
| GenBot                                                      | Genetic algorithms | (Liu et al., 2017)             | Improved results of state-of-the-art techniques                           | Cresci-17                                                                                                                                                                  | User information and content features.                                         |
| LA-SMBD & trust model & Bayesian learning & Dempster-Shafer | Hybrid             | (Rout et al., 2020)            | Precision: 95.37%                                                         | Cresci-2015 Social HoneyPot                                                                                                                                                | User information, content, network, and temporal features                      |
| S3MCD: ADMM & LR                                            | Hybrid             | (Wu et al., 2016)              | Recall: 93.08%<br>Precision: 92.34%<br>F1-score: 92.71%                   | Scraped dataset from Twitter                                                                                                                                               | User information, content, and friend features                                 |
| Similarity-based MRF and RF                                 | Hybrid             | (Al-Rakhami and Al-Amri, 2020) | Accuracy 95.2%                                                            | Scraped dataset from Twitter                                                                                                                                               | User information, content network, and behavioral features                     |
| Social Fingerprinting: Digital DNA model                    | DNA-inspired       | (Cresci et al., 2017)          | Precision: 98.2%<br>Accuracy: 97.7%<br>F-measure: 97.7%<br>Precision: 94% | Retweeters of the political candidate Spammers of Amazon.com products<br>Cresci-2015                                                                                       | Behavioral features                                                            |
| DEBOT & Warped Correlation                                  | Hybrid             | (Wei and Nguyen, 2019)         | AUC: 99.56%                                                               | Scraped dataset from Twitter                                                                                                                                               | Content features                                                               |
| Contrast model, DT, RF                                      |                    | (Loyola-Gonzalez et al., 2019) |                                                                           | Cresci-17 dataset                                                                                                                                                          | User information, content, temporal, and sentiment features                    |
| Big data and RDF                                            |                    | (Caruccio et al., 2018)        | –                                                                         | Cresci-2015                                                                                                                                                                | User information                                                               |
| KNN & HawkesEye & Topic modeling                            | Hybrid             | (Dutta et al., 2020)           | Recall: 96%                                                               | Verified and Real account dataset<br>Scraped dataset from Twitter                                                                                                          | User information, content, and temporal features                               |
| StreamKM ++ & DenStream                                     | Anomaly-based      | (Miller et al., 2014)          | Recall: 100%<br>TF: 2.2%                                                  | Scraped dataset from Twitter                                                                                                                                               | User information, content features                                             |
| SVM-NN                                                      | Hybrid             | (Khaled et al., 2018)          | Accuracy: 98.3%                                                           | The Cresci-2015 dataset; Dataset, human accounts: human of politicians, journalists and bloggers, Fastfollowerz dataset., InterTwitter dataset, Twitter technology dataset | User information                                                               |
| Adaboost and Topic Modeling                                 | Hybrid             | (Morstatter et al., 2016)      | Recall: 82.48%<br>Precision: 75.41%                                       | Libya dataset, Arab spring activity in Libya.<br>Arabic HoneyPot                                                                                                           | Content features                                                               |
| random walk & Deep Autoencoder                              | Hybrid             | (Lingam et al., 2020)          | Precision: 90%                                                            | Scraped dataset from Twitter                                                                                                                                               | Content, behavioral features                                                   |
| Entropy, LDA, Bayesian algorithm                            | Hybrid             | (Chu et al., 2010)             | TP: 82.80%                                                                | Scraped dataset from Twitter                                                                                                                                               | User information, content features                                             |
| LDA                                                         | Topic modeling     | (Koggalahewa et al., 2022)     | Accuracy: 96.9%                                                           | Dataset of Twitter user accounts posting about fashion                                                                                                                     | Content features                                                               |
| BotCamp: boosted TD, graph clustering                       | Hybrid             | (Abu-El-Rub and Mueen, 2019)   | Accuracy: 87.5%<br>Precision: 98%                                         | Scraped dataset from Twitter                                                                                                                                               | Content, temporal, and network features                                        |
| LDA & LCS                                                   | Topic modeling     | (Liu et al., 2014)             | Precision: 99%                                                            | real spam dataset scraped from Sina Weibo                                                                                                                                  | User information, content, and network features                                |

**Table 7**  
Online Social Networks Features.

| Features                               | Description                                                                                                                                                                                                                    |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User information                       | features and <i>meta</i> -data extracted from users' profiles, including username or screen name, language, bio, etc.                                                                                                          |
| Friend                                 | Include statistics and numerical attributes related to the user's friends or followers, such as the median, the number of followers, the number of accounts followed, the follower ratio, etc.                                 |
| Temporal features                      | They are employed to discover the existence of synchronized activities on OSN. Examples are the average time between two successive publications, the time between two consecutive retweets, etc.                              |
| Content                                | The bot-generated contents typically use simple terminology and short clauses.                                                                                                                                                 |
| Sentiment                              | Features computed to capture the polarity of the content posted in OSN and the posture of an integral conversation based on emotions score, polarity score, n-grams features, etc.                                             |
| Network features                       | Features that rely on users' social interactions and activities and capture patterns of the social network of a community or a group of accounts.                                                                              |
| Contextual features or deep linguistic | Features that capture the aspects of the construction of tweets and conserve the semantic or syntactic form of tweets using Deep learning and NLP techniques.                                                                  |
| Behavioral features                    | Features that describe the action and interactions of social bots in OSN. So, the behavioral similarity can be examined by comparing content similitude, URL co-occurrence in posts, social activity, and reaction similarity. |

**Fig. 9.** Features classes used to detect each type of SMB.

For the DL algorithms, we observe that they are less used compared to traditional ML algorithms. Therefore, as shown in Fig. 8, the most commonly used algorithms are CNN, LSTM, ANN, and BI-LSTM. These algorithms were applied for the generic detection of SMBs, spam bots, and Sybil. To assess the performance of the SMB detection approaches based on DL techniques, we summarize in Table 4 the DL algorithms carried out and their highest performance related to the dataset and the features used to train the model.

To develop models to find SMBs, researchers have used a variety of DL algorithms. As shown in Fig. 8, the LSTM, CNN, and ANN models are the most commonly used algorithms. Based on the results presented in Table 4, we note that LSTM and CNN are the best-performing algorithms. Both used content features and user information to achieve 100% recall and 100% accuracy on the Cresci-17 dataset. In addition, we observed that user information, content, network, and temporal features are the most common characteristics used for DL-based detection systems. The methods based on graph models and graph theory enable building the graph structure of the OSN based on the connections and relationships among the OSN nodes. Generally, it uses graph theory to rank OSN accounts, which provides a robust mechanism for identifying SMBs and detecting trustworthy users. In other works, they used DL algorithms to detect the botnet and SMB community. Bots often form clusters or networks and may engage in coordinated behavior.

Analyzing the network structure can help identify bot accounts. Table 5 gives a global view of the results of graph-based algorithms used in the studied works.

The majority of approaches for the detection of fake accounts are based on mathematical algorithms from graph theory. It is possible to build the social graph of the studied accounts through this algorithm. In this case, the nodes are the accounts, and the edges are the social links between the accounts. To build the structure of the social graph of the studied accounts, it is necessary to have additional information such as the network features, the list of friends, the set of interactions between users, etc. Combining user information and network features, the SybilBelief approach (Gong et al., 2014) achieved 100% accuracy in detecting Sybil nodes. Then comes Sybil-Resist (Ma et al., 2014); which earned a detection rate of 99.2% to detect the Sybil region and nodes based on the Random Walk algorithm.

Many other techniques were used for SMB detection based on genetic-based algorithms, digital DNA-based algorithms, anomaly-based algorithms, topic modeling-based approaches, and hybrid approaches. Table 6 presents the findings of each of these algorithms:

The anomaly-based approaches (Davis et al., 2016); (Miller et al., 2014); (Rodríguez-Ruiz et al., 2020) used Clustering algorithms to identify OSNs intrusions and attacks. Therefore, the topic modeling techniques were employed to detect active SMBs in

emerging topics in OSNs. The highest precision achieved by the topic-modeling-based technique is 99% (Liu et al., 2014). The approaches based on DNA-inspired and Genetic algorithms were used to detect spam bots, reaching a precision of over 95%. The hybrid approaches proved that combining several algorithms increased the efficiency of the malicious SMBs recognition, and the precision is up to 95% which is very promising. To improve the performance of the SMB detectors, it's crucial to continually update the models with new and different types of content to ensure that they can recognize and block the latest tactics used by SMBs and other spammers. This requires ongoing training and testing of the models to ensure they are up-to-date and effective at detecting and blocking SMBs.

#### 4.3. Feature types in social bot detection systems (RQ3)

Characterizing the relevant features of public social data is a critical step in detecting social bots. Several feature categories have been investigated, but six primary feature classes have been identified as important in distinguishing between human users and SMBs. We propose a new categorization of OSN features in this paper (Ellaky et al., 2021): user information features, friend features, temporal features, content features, sentiment features, network/interaction features, deep linguistic features, and behavioral features, as defined in Table 7.

All the SMB detection techniques under study were built using a set of OSN account attributes. Based on the feature mentioned above categories, we evaluated the feature utilization rate to determine each type of SMB. It is challenging to compare how frequently each feature type is employed in identifying each type of bot because we have a different number of articles in each category. Yet, it demonstrates the most common characteristics used to classify various social bots. Nonetheless, it would be intriguing to investigate whether the effectiveness of the detection systems examined in this paper correlates with the employed features. Fig. 9 illustrates the OSN features used to detect each type of SMB.

From Fig. 9, we can observe that all research studies on social bots' detection used content features, user information, network features, and behavioral features. We also noticed that the deep linguistic and contextual features had not been carried out enough to detect social bots. The same feature was the most used for Sybil and cyborgs' detection. All techniques investigated in the context of stegobots focused solely on content features and user information. Therefore, the content feature, user information, and network features were also the most commonly used for the spam bot. Indeed, for political bots' detection systems, the most used features

are content, user information, temporal features, network features, friend features, and sentiment features.

#### 4.4. Feature engineering and preprocessing techniques in social bot detection systems (RQ4)

OSN metadata includes numerous information associated with contents and events, such as the time and location from where a post was sent. To train a model using OSN-based datasets that contain misspellings, informal language, and symbolic phrases, preprocessing is necessary to process and analyze the data, as well as feature extraction and selection.

##### 4.4.1. Preprocessing techniques

The content feature represents the comments posted by OSN users. The comments are written unconventionally and include text, hashtags, numbers, emoticons, etc. Natural Language (NLP) techniques are required to clean and preprocess text data. Preprocessing text data helps train a model with relevant data to keep the pattern required to build accurate ML models (Ellaky et al., 2021).

##### • Cleaning

Cleaning the data is necessary to make the dataset available in a computational format without changing its meaning or content. This includes removing elongations, diacritics, punctuation, digits,

**Table 9**  
Feature selection techniques.

| Techniques                      | References                                                                                                                                                                 |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manual                          | (Pratama and Rakhmawati, 2019) (Rodríguez-Ruiz et al., 2020)                                                                                                               |
| Correlation measure             | (Natarajan et al., 2015) (Khaled et al., 2018) (Lingam et al., 2018) (Fernandes et al., 2015)                                                                              |
| Entropy values                  | (Ji et al., 2016)                                                                                                                                                          |
| IG                              | (Kang et al., 2016) (Kantepe and Ganiz, 2017)                                                                                                                              |
| l1-regularized linear SVM       | (Bello et al., 2018)                                                                                                                                                       |
| MI, Chi Square                  | (Kantepe and Ganiz, 2017)                                                                                                                                                  |
| PCA                             | (Alsaleh et al., 2014) (Beğenilmiş and Uskudarli, 2018) (Al-Qurishi et al., 2018) (Dickerson et al., 2014) (Gilani et al., 2019) (Khaled et al., 2018) (Lee and Kim, 2014) |
| SVM                             | (Khaled et al., 2018)                                                                                                                                                      |
| CVAE                            | (Alothali et al., 2021)                                                                                                                                                    |
| Wrapper subset evaluation (WSE) | (Alothali et al., 2021)                                                                                                                                                    |
| Filter                          | (Alothali et al., 2021)                                                                                                                                                    |
| Embedded                        | (Alothali et al., 2021)                                                                                                                                                    |

**Table 8**  
Feature extraction techniques.

| Techniques      | References                                                                                                                                                                                    |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N-gram          | (Miller et al., 2014) (Beskow and Carley, 2019) (Bello et al., 2018) (Inuwa-Dutse et al., 2018) (Lee and Kim, 2014) (Almerikhi and Elsayed, 2015) (Ikram, et al., 2017) (Ellaky et al., 2021) |
| TF-IDF          | (Beskow and Carley, 2019) (Ferrara, 2017) (Luceri et al., 2019) (Ellaky et al., 2021) (Wu et al., 2016)                                                                                       |
| BOW             | (Ellaky et al., 2021)                                                                                                                                                                         |
| BERT            | (Ellaky et al., 2021) (Kumar et al., 2021) (Grinberg et al., 2016)                                                                                                                            |
| LDA             | (Morstatter et al., 2016) (Bello et al., 2018) (Al-Qurishi et al., 2018)                                                                                                                      |
| LSA             | (Inuwa-Dutse et al., 2018) (Chew et al., 2018)                                                                                                                                                |
| GloVe           | (Benabbou et al., 2022)                                                                                                                                                                       |
| Roberta         | (Martin-Gutierrez et al., 2021)                                                                                                                                                               |
| Word2vec        | (Ellaky et al., 2021)                                                                                                                                                                         |
| Fastext         | (Ellaky et al., 2021)                                                                                                                                                                         |
| Doc2vec         | (Ellaky et al., 2021)                                                                                                                                                                         |
| Rule Extraction | (Bello et al., 2018)                                                                                                                                                                          |
| Hybrid: LR-PCA  | (Murugan and Devi, 2019)                                                                                                                                                                      |
| Manual          | (P. G. Pratama et N. A. Rakhmawati, , 2019)                                                                                                                                                   |



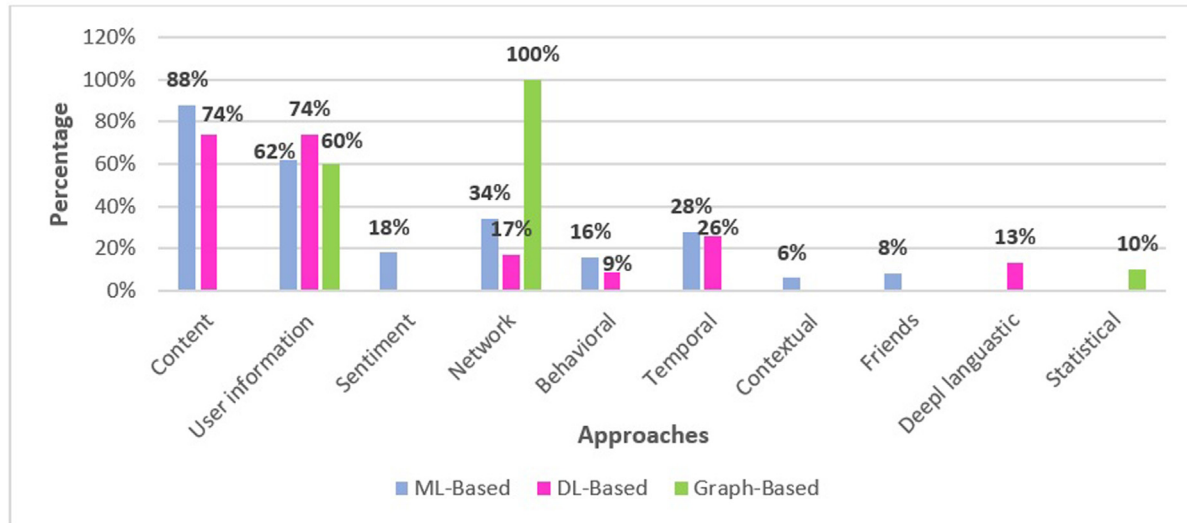


Fig. 10. Features used on the ML-Based, DL-based, and Graph-based algorithms.

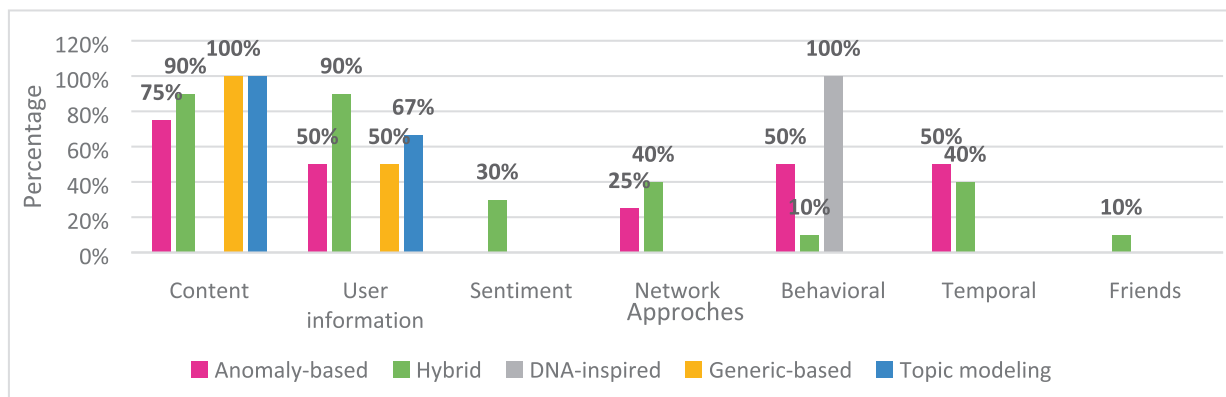


Fig. 11. Features used on the Anomaly-Based, DNA-inspired, Genetic-based, Topic-modeling and Hybrid approaches.

and superfluous characters. Therefore, most of the studied works followed a standard NLP cleaning process. Several cleaning methods have been employed since they reduce noise and enhance NLP comprehension. The elimination of undesirable characters, such as non-Arabic or non-English characters, punctuation, stop words, motion (@), hashtags (#), repeated letters, letter lengthening, diacritical marks, URLs, and usernames (Bello et al., 2018) (Almerekhi and Elsayed, 2015) (Ping and Qin, 2018) (Andriotis and Takasu, 2018). However, some removed symbols or characters can be discriminant for bots' detection because it could result in the loss of essential data. It would be interesting, for example, to calculate the number of URLs, motions, and hashtags, replacing emojis and emoticons with meaningful words and replacing acronyms with their complete form.

#### • Stop words

The majority of authors eliminate stop-words by utilizing libraries that contain words explicitly compiled that are used for text preprocessing or by employing stop words because they are a collection of keywords that are irrelevant for analysis like for the Arabic content that can be automatically preprocessed it

requires information retrieval systems (IR) to retrieve the irrelevant words to eliminate during the preprocessing phase. Other techniques, including automated and hybrid (semi-automatic) techniques, have been applied in a few works.

#### 4.4.2. Feature extraction

The datasets used for SMB detection contain several types of features, for example, textual data, numerical data, images, etc. The initial dataset requires a computation phase to calculate the numerical and statistical features and convert the unstructured attributes into a computer-readable format. Feature extraction is a method employed to decrease the dimensionality space of data to a compact dimensionality space to get manageable data. The new features are generated by combining and transforming the original features. Applying feature extraction techniques permits to increase in the training model's performance and decreases the learning process's cost (Safaa et al., 2022). In literature, many techniques were applied for feature extraction, such as statistical methods are used to extract statistical and numerical features from user information, contents, and network features (ex: following/follower ratio, URL ratio, medium frequency of sharing, etc.) (Al-Rakhami and Al-Amri, 2020); (Balestrucci et al., 2021), LDA

**Table 10**  
Efficient features for SMBs detection.

| Social Bot Type | Features                                                                                                                            |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| SMBs            | Content: 78%, user information:62%, network features: 40%, temporal features:26%                                                    |
| Sybil, cyborg   | Content: 33%, user information:53%, behavioral features: 53%, temporal features:20%                                                 |
| Stegobot        | Content: 100%, user information:100%, behavioral features: 50%, network features:50%                                                |
| Spam bots       | Content: 91%, user information: 70%, behavioral features: 13%, network features:40%, sentiment features: 13%, temporal features:13% |
| Political bots  | Content: 60%, user information: 80%, network features:60%, sentiment features: 40%, temporal features: 80%                          |
| Game bots       | Content: 100%, user information: 100%, behavioral features: 100%                                                                    |

(Bello et al., 2018); (Dutta et al., 2020); (Liu et al., 2014), and manual feature extraction (Pratama and Rakhmawati, 2019). The LSA, and TF-IDF are used to represent text-based content like tweets and comments. Table 8 provides an overview of all feature extraction techniques used in the state-of-the-art works.

From the state-of-the-art studied techniques, the most used techniques for feature extraction are N-gram and TF-IDF. The LDA and LSA methods are typically used for topic modeling the content used in the case of political bots' detection to determine the most important topic discussed in OSNs. However, some embedding techniques have been investigated, such as Bert, GloVe, Word2vec, Doc2vec, Roberta, and Fastext. In this research work (Ellaky et al., 2021), the efficiency of different embedding methods was compared on ML-based models to detect SMBs. The experiment results were very promising, using RF and DT algorithms and Doc2Vec, which reached a precision score of 100%. Certain techniques, such as DistilBERT, XLNet, and sent2vec, have not been applied enough in the literature, which are achieving excellent performance on various NLP tasks, such as question answering, text classification, and sentiment analysis. DL has enabled a boost in performance for NLP applications. It is superior to traditional ML algorithms due to their single end-to-end architecture and ability to handle large training data. However, several works on deep learning and NLP for SMB detection have been published since 2019. Researchers have still not applied many of these techniques to improve SMB detection.

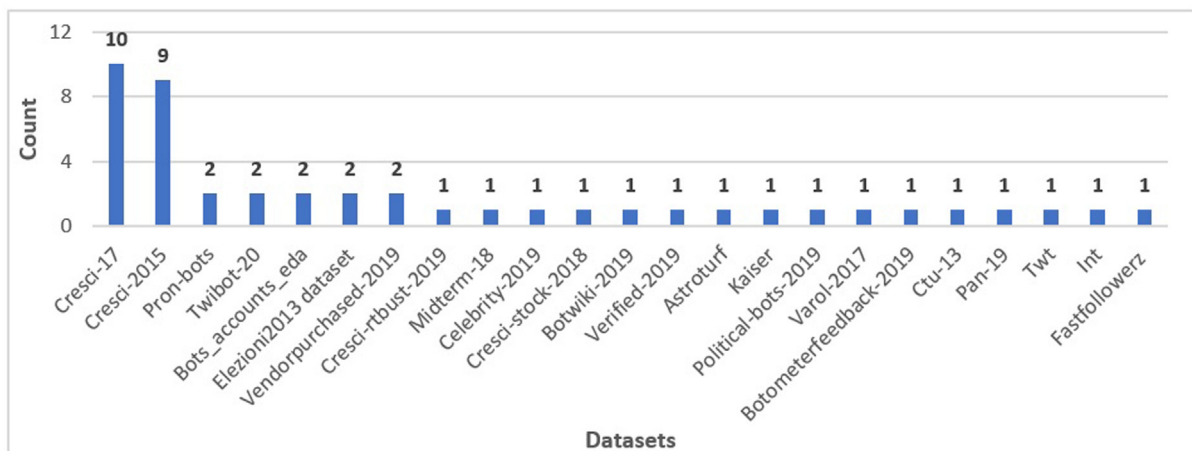
#### 4.4.3. Feature selection

Using selection techniques helps to evade the curse of dimensionality phenomena. It minimizes the dimension of data sets by selecting a limited subset of features that efficiently define the data (Safaa et al., 2022). From the state of the literature, we observed that many feature selection techniques were used. In this work (Pratama and Rakhmawati, 2019), the authors used manual feature selection to detect political bots. In (Rodríguez-Ruiz et al., 2020), the authors manually selected behavioral and numeric features to detect SMBs based on one class classifier. In this work, the authors used (Allothali et al., 2021) information gains ranking filters to avoid overfitting and to reduce the features. And in this work, the authors used Information Gain (IG), Mutual Information (MI), and Chi-Square (Kantepe and Ganiz, 2017). Moreover, in this study (Beğenilmiş and Uskudarli, 2018); the authors used PCA for feature selection to detect organized behavior based on Tweets. The Wrapper methods (Feng et al., 2019) use the model's performance to evaluate the selected feature set. Table 9 presents the feature selection techniques used in the state-of-the-art.

The most common feature selection method used by researchers is PCA. It reduces the number of features by choosing the dimensions of features that exhibit the most significant variation (Murugan and Devi, 2019), thereby enhancing the performance of SVM-NN (Khaled et al., 2018), which can identify fake accounts with 98.3% accuracy and fewer features.

**Table 11**  
Dataset sources usage.

|                   | Twitter | Facebook | Sina Weibo | Renren | Flickr | YouTube | Reddit | VKontakte |
|-------------------|---------|----------|------------|--------|--------|---------|--------|-----------|
| Number of Studies | 113     | 7        | 4          | 4      | 3      | 1       | 2      | 1         |

**Fig. 12.** Public datasets used for social bots' detection.

**Table 12**  
Public datasets provided for SMBs detection.

| References                                  | Name                    | Description                                                                                                      | Size                                                                       | Citation                                                                                                                                                                                                                                                                       |
|---------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (Yang et al., 2019)                         | Pron-bots               | It contains Twitter bot accounts and user information.                                                           | Bots: 21963                                                                | (Feng et al., 2021) (Sayyadiharikandeh et al., 2020)                                                                                                                                                                                                                           |
| (Cresci et al., 2017) (Cresci et al., 2017) | Cresci-17               | Dataset of genuine, traditional, and social spam bots.                                                           | Users: 2764, spam bots: 7049                                               | (Mohammad et al., 2019) (Feng et al., 2021) (Ellaky et al., 2021) (Kumar et al., mai 2021) (Benabbou et al., 2022) (Andriotis and Takasu, 2018) (Cresci et al., 2019) (Loyola-Gonzalez et al., 2019) (Wei and Nguyen, 2019) (Rodríguez-Ruiz et al., 2020) (Mazza et al., 2019) |
| (Mazza et al., 2019)                        | Cresci-rtbust-2019      | Dataset of the bot and human accounts.                                                                           | Users: 693                                                                 |                                                                                                                                                                                                                                                                                |
| (Pizarro, 2020)                             | Midterm-18              | Dataset of the politically committed users.                                                                      | Users: 50538                                                               | (Pizarro, 2020)                                                                                                                                                                                                                                                                |
| (Yang et al., 2019)                         | Celebrity-2019          | Dataset of celebrity accounts.                                                                                   | Users: 5971                                                                | (Yang et al., 2019)                                                                                                                                                                                                                                                            |
| (Igawa, et al., 2016)                       | Cresci-stock-2018       | Dataset of Twitter accounts posting about fashion.                                                               | Users: 13276                                                               | (Miller et al., 2014)                                                                                                                                                                                                                                                          |
| (Cresci et al., 2018)                       | Botwiki-2019            | Dataset of bots from <a href="https://botwiki.org">https://botwiki.org</a> .                                     | Bots: 698                                                                  | (Cresci et al., 2018)                                                                                                                                                                                                                                                          |
| (Yang et al., 2020)                         | Verified-2019           | Dataset of Twitter verified accounts.                                                                            | Users: 1987                                                                | (Yang et al., 2020)                                                                                                                                                                                                                                                            |
| (Feng et al., 2021)                         | Twibot-20               | Dataset of the recent generation of Twitter bots and genuine users.                                              | User: 229573, tweets: 33,48 M, followers: 455,958                          | (Feng et al., 2021) (Feng et al., 2021)                                                                                                                                                                                                                                        |
| (Sayyadiharikandeh et al., 2020)            | Astroturf               | Dataset of hyperactive politicals.                                                                               | Bots: 505                                                                  | (Ma et al., 2014)                                                                                                                                                                                                                                                              |
| (Rauchfleisch and Kaiser, 2020)             | Kaiser                  | Dataset of German bots' official accounts, German members of parliament, and members of the 115th U.S. Congress. | German bots: 27, official accounts: 532, parliament accounts: 516          | (Rauchfleisch and Kaiser, 2020)                                                                                                                                                                                                                                                |
| (Yang et al., 2019)                         | Political-bots-2019     | Dataset of bots run by @rzazula), shared by @josh_emerson.                                                       | Bots: 62                                                                   | (Yang et al., 2019)                                                                                                                                                                                                                                                            |
| (Varol et al., 2017)                        | Varol-2017              | Dataset of human and bot accounts.                                                                               | Bots: 1525, humans: 451717                                                 | (Varol et al., 2017)                                                                                                                                                                                                                                                           |
| (Cresci et al., 2015)                       | Cresci-2015             | Dataset of Fake Twitter accounts.                                                                                | Users: 3900, tweets: 2.75 M, followers: 1.81 M, friends: 1.78 M            | (Rout et al., 2020) (Pham et al., 2022) (Mulamba et al., 2016) (Khaled et al., 2018) (Caruccio et al., 2018) (Lingam et al., 2018) (Cresci et al., 2019) (Lingam et al., 2021) (Zhao et al., 2020) (Yang et al., 2019) (Sayyadiharikandeh et al., 2020)                        |
| (Yang et al., 2013)                         | Vendor-purchased-2019   | Dataset of fake follower accounts.                                                                               | Users: 1088                                                                |                                                                                                                                                                                                                                                                                |
| (Yang et al., 2019)                         | Botometer-feedback-2019 | Dataset of human and bot accounts.                                                                               | Users: 386, bots: 143                                                      | (Yang et al., 2019)                                                                                                                                                                                                                                                            |
| (García et al., 2014)                       | CTU-13 dataset          | Dataset of botnet traffics.                                                                                      | –                                                                          | (J. Wang et I. Ch. Paschalidis, « Botnet Detection Based on Anomaly and Community Detection », IEEE Trans. Control Netw. Syst., vol. 4, no 2, p. 392 404, juin, 2017) (Feng et al., 2021)                                                                                      |
| (Pizarro, oct. 2020)                        | Pan-19                  | Dataset of SMBs and gender profiling.                                                                            | English: human: 2060, human-tweets: 206000, bots: 2060, bot-tweets: 206000 |                                                                                                                                                                                                                                                                                |
| (Dorri et al., 2018)                        | Bots_accounts_eda       | Twitter accounts of humans and bots.                                                                             | Users: 25013, bots: 12,425                                                 | (Dorri et al., 2018) (Lingam et al., 2018)                                                                                                                                                                                                                                     |
| (Cresci et al., 2015)                       | Elezioni2013 dataset    | Fake accounts dataset.                                                                                           | Accounts: 1481, tweets: 2.06 M, followers: 1.52 M, friends: 667225         | (Mulamba et al., 2016) (Danezis and Mittal, 2009)                                                                                                                                                                                                                              |
| (Cresci et al., 2015)                       | TWT                     | Fake accounts datasetS                                                                                           | Accounts: 845, tweets: 114192, followers: 28588, friends: 729839           | (Mulamba et al., 2016)                                                                                                                                                                                                                                                         |
| (Cresci et al., 2015)                       | INT                     | Fake accounts dataset.                                                                                           | Accounts: 1337, tweets: 58925, followers: 23173, friends: 517485           | (Mulamba et al., 2016)                                                                                                                                                                                                                                                         |
| (Danezis and Mittal, 2009)                  | Fastfollowerz           | Fake accounts dataset.                                                                                           | Accounts: 1169                                                             | (Danezis and Mittal, 2009)                                                                                                                                                                                                                                                     |

#### 4.4.4. Features used in the detection algorithms

This section aims to highlight the features the researchers have exploited to build their detection systems and understand if there is a strong relationship between the selected features and the type of bot detected. We also investigate the impact of feature selection on the design of efficient models for SMB recognition. Fig. 10 illustrates the percentages of the different feature types used to design the ML-Based, DL-Based, and Graph-Based methods.

Fig. 10 shows that the most used features for ML-based systems are content, user information, network, and temporal features. In the case of DL-based algorithms, the most used features are content, user information, network, temporal fea-

tures, and deep linguistic features. Hence, there is a great deal of similarity between ML and DL-based approaches. Furthermore, DL-based algorithms take advantage of DL's evolution in the NLP domain, which has enabled the use of deep linguistic features such as embedding techniques. Graph-based algorithms' most commonly used features are user information, network, and statistical features. Indeed, they use these features to build the graph, which represents the structure of the investigated accounts' social network and allows the detection of the group of SMBs that engage in malicious activities in OSNs. Fig. 11 plots the percentages of the varying feature types used to implement the other detection approaches.

**Table 13**  
Detail of the Cresci datasets.

| Dataset Name | Content                    | Description                                                                                          | Size                                                                          | Features                                                |
|--------------|----------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------|
| Cresci-17    | D1: genuine accounts       | Human accounts                                                                                       | Account: 3474, Tweets: 8377, Year: 2011                                       | User information, content, friend, and network features |
|              | D2: Social spam bots1      | Re-tweeters of Italian political candidate                                                           | Account: 991, Tweets: 1.61 M, Year: 2012                                      |                                                         |
|              | D3: Social spam bots2      | Spammers of paid applications for mobile                                                             | Account: 3457, Tweets: 428542, Year: 2014                                     |                                                         |
|              | D4: Social spam bots3      | Spammers of products on sale at <a href="https://www.amazon.com">Amazon.com</a>                      | Account: 464, Tweets: 1.41 M, Year: 2011                                      |                                                         |
|              | D5: Traditional spam bots1 | Spammers                                                                                             | Account: 1000, Tweets: 145094, Year: 2009                                     |                                                         |
|              | D6: Traditional spam bots2 | Spammers using scam URLs                                                                             | Account: 100, Tweets: 74957, Year: 2014                                       |                                                         |
|              | D6: Traditional spam bots3 | Spammers of job offer                                                                                | Account: 433, Tweets: 5.79 M, Year: 2013                                      |                                                         |
|              | D7: Traditional spam bots4 | Spammers of job offers                                                                               | Account: 1128, Tweets: 133311, Year: 2009                                     |                                                         |
|              | D8: Fake followers         | Accounts of fake followers                                                                           | Account: 3351, Tweets: 196027, Year: 2012                                     |                                                         |
| Cresci-2015  | D1: ThefakeProjetct        | Verified human accounts                                                                              | Account: 469, Tweets: 563693, Followers: 258494, Friends: 241,710             | User information, content, friend, and network features |
|              | D2: elezioni13             | Verified human accounts                                                                              | Account: 1481, Tweets: 2.06 M, Followers: 1.52 M, Friends: 667225, Year: 2013 |                                                         |
|              | D3: fastfollowerZ          | Fake accounts from <a href="https://www.fastfollowerz.com">https://www.fastfollowerz.com</a>         | Account: 1169, Tweets: 22910, Followers: 11893, Friends: 253026               |                                                         |
|              | D4: interTwitter           | Fake accounts from <a href="https://www.interTwitter.com">https://www.interTwitter.com</a>           | Account: 1337, Tweets: 58625, Followers: 23173, Friends: 517485               |                                                         |
|              | D5: Twittertechnology      | Fake accounts from <a href="https://www.Twittertechnology.com">https://www.Twittertechnology.com</a> | Account: 845, Tweets: 114192, Followers: 28588, Friends: 729839               |                                                         |
|              | D6: human                  | Verified human accounts                                                                              | Account: 1950, Tweets: 2,63 M, Followers: 1,78 M, Friends: 908935             |                                                         |
|              | D7: Fake                   | Fake Twitter accounts                                                                                | Account: 1950, Tweets: 118327, Followers: 34553, Friends: 879580              |                                                         |
|              | D8: Baseline               | Union of D6 and D7                                                                                   | Account: 3900, Tweets: 2.75, Followers: 1.81 M, Friends: 1.78 M               |                                                         |

**Table 14**  
Sampling Techniques.

| Techniques                                                           | References                             |
|----------------------------------------------------------------------|----------------------------------------|
| Random Oversampling, CGAN, SMOTE, ADASYN                             | (Wu et al., 2020)                      |
| SMOTE                                                                | (Wu et al., 2020) (Fazil et al., 2021) |
| Snowball sampling                                                    | (Beskow and Carley, 2018)              |
| Entropy-based uncertainty sampling                                   | (Wu et al., 2021)                      |
| Synthetic minority oversampling                                      | (Kudugunta and Ferrara, 2018)          |
| Cost-sensitive learning                                              | (Zhao et al., 2020)                    |
| Fuzzy-based oversampling, random oversampling, random under sampling | (Liu et al., août 2017)                |

From Fig. 11, we observe that the most utilized features in anomaly-based systems are content, user information, network, and temporal features. In DNA-based systems, the most frequently employed attributes are behavioral features. Additionally, the most used features for genetic-based and topic-modeling-based systems are content and user information. Conversely, we noticed that the most used features for hybrid models are content, user information, network, temporal, and sentiment features. These results are also confirmed by Table 4, Table 5, and Table 6, which showed that user information, content, network, and temporal features are the attributes used in the ML and DL models, resulting in the best performance. In Table 10, we summarized the most valuable features to detect each type of SMB to give some insights to researchers.

As a result, the content, user information, and network features are the most commonly used attributes to conduct a specific or generic detection of SMBs. Even if they have not been exploited

much, sentiment features proved to be effective in detecting political bots that aim to influence the opinion of OSN users by posting positive or negative content. In addition, the temporal features have proven their effectiveness in detecting synchronized massive activities related to events like the presidential elections. Therefore, behavioral features are commonly used to detect Sybil, Cyborg, and game bots.

#### 4.5. Dataset issues (RQ5)

SMB detection systems face several problems related to the dataset characteristics. Indeed, some of these issues have a direct impact on the performance of the approach. The topics addressed in the state-of-the-art include the problem of dataset size, unbalanced data, and data labeling. Furthermore, social bots keep changing their characteristics and behaviors to escape detection systems which require new datasets that follow the evolution of the social



**Table 15**  
Dataset labeling techniques.

| Technique            | Strengths and Limits                                                                                                                                                                                                                                                           | References                                                                                                                                                                                                                                                               |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manual               | 1) Labeling process takes a long time; 2) Not efficient because of human error; 3) Small set of data is produced, the labeling task becomes complex when the size of the data increases; 4) It does not take into consideration the behavior of the bot during the annotation. | (Fazil et al., 2021) (Bello et al., 2018) (Danezis and Mittal, 2009) (Yang et al., 2019) (Cresci et al., 2016) (Gilani et al., 2019) (Cresci et al., 2017) (Shetty et al., 2021) (Chew et al., 2018) (Jeong et al., 2016) (Safaa et al., 2022) (AlRubaiyan et al., 2015) |
| Suspended users list | 1) Simple technique; 2) Depends on the accuracy of the detection system used by the OSN and criteria used for the suspension.                                                                                                                                                  | (Dickerson et al., 2014) (Ping et S. Qin, 2018)                                                                                                                                                                                                                          |
| Honeypots            | Permit to collect a large set of active bots with considerable reliability.                                                                                                                                                                                                    | (Dickerson et al., 2014) (Koggalaheewa et al., 2022)                                                                                                                                                                                                                     |
| Semi-automatic       | Rapid labeling of large datasets enables the quick implementation and adjustment of models to new settings and contexts.                                                                                                                                                       | (Beskow and Carley, 2019) (Inuwa-Dutse et al., 2018)                                                                                                                                                                                                                     |
| Automatic            | 1) Simple and rapid labeling; 2) Labeling efficiently depends on the used API. The most commonly used detection tools are Botometer (Davis et al., 2016), Botometer101 (Yang et al., 2022), and Debot (Chavoshi et al., 2016).                                                 | (Bruno et al., 2022) (Abu-El-Rub and Mueen, 2019)                                                                                                                                                                                                                        |

bots for effective detection systems. The authors used datasets scraped from OSN by utilizing APIs (Dickerson et al., 2014); (Ji et al., 2016); (Jr et al., 2018) such as Tweepy and Twitter API in some works. On the other hand, many public datasets are suggested by researchers for SMB detection, and they are available for download in Bot Repository (<https://botometer.osome.iu.edu/bot-repository/datasets.html>).

#### 4.5.1. Dataset sources

Twitter is the most extensively utilized source for data collecting and dataset building for training ML and DL algorithms. Twitter has a data feed API that allows users to collect data in real-time and provides researchers full access to the user's data that need to analyze SMBs and establish detection systems in place. A statistical breakdown of the various existing OSNs' utilization rates is shown in Table 11. As we can see, less research has been done on Facebook, Sina Weibo, Renren, and Flickr than on Twitter.

For spam bots, Sybils, and cyborgs, most researchers used public datasets or combined several datasets to have a dataset corresponding to the model requirements regarding size, diversity of data, features, and data scarcity. For the detection of political bots, in most cases, data was collected using the hashtags of candidates in the electoral elections. Fig. 12 provides the count distribution of public datasets used to detect SMBs.

As shown in Fig. 12, many public datasets address the topic of SMB recognition, and they have been used to train and evaluate numerous state-of-the-art research projects. We identified 23 public datasets devoted to SMB detection. The most used datasets are Cresci-17 and Cresci-2015. For more details, Table 12 gives a presentation of the collected datasets as follows:

- Column References: refers to the article in which the dataset was proposed.
- Name: is the name of the dataset.
- Description: provides a description of the dataset and the type of SMBs it contains.
- Size: is the size of the dataset.
- Citation: refers to the works that used the dataset for their experimentation.

Researchers widely use the Cresci-17 and the Cresci-2015 because they contain various features, which provide more opportunities for researchers to investigate different methods using different characteristics. Since they are labeled datasets, they are used for training supervised models based on ML, DL, genetic, DNA, graph algorithms, etc. The Cresci-17 dataset provides a large set of genuine, traditional, spammer, and fake follower accounts extracted from Twitter, labeled manually by CrowdFlower collaborators by investigating the attributes of each account. The Cresci-

2015 dataset includes genuine and fake accounts extracted from Twitter and manually labeled based on the attributes of the scraped data. Table 13 provides more details about these datasets.

It is evident from Table 5 that the Cresci-2015 and the Cresci-17 dataset provide a large set of data and features. They gave good results for approaches based on ML, DL, graph theory, DNA-inspired algorithms, anomaly-based, etc. On the other hand, these datasets contain data that is considerably old. It does not follow the evolution of SMBs and omits the current active SMBs on OSNs. Furthermore, it does not cover the diversity of SMBs each dataset focuses on a specific type of social bot.

#### 4.5.2. Imbalanced data

Another challenge with datasets is the imbalanced data problem. In imbalanced datasets, data may have an asymmetry in the distribution from one class to another. This situation may bias the classification results. In addition, the impact of the choice of sampling techniques on the performance of DL models is significant, and often researchers investigate several methods before making a final choice. For this aim, some techniques were applied, whether under-sampling, over-sampling, or hybrid methods. Table 14 presents the sampling techniques used in the studied articles.

Table 14 shows that only some works have dealt with the problem of imbalanced data, even though the datasets used in the state-of-the-art papers are all imbalanced. However, the proportion of legitimate accounts is superior to the proportion of SMBs. Indeed, training the model with an imbalanced dataset will lead to a biased model that overfills the majority class. Training the model using balanced data has increased the accuracy of spam bot detection in (Fazil et al., 2021) with an accuracy rate of 99.83%, in (Kudugunta and Ferrara, 2018) with an accuracy of 96%, and in the other works referenced in Table 14.

#### 4.5.3. Data labeling

One of the significant challenges facing researchers in the field of SMB detection is to have labeled datasets for training supervised ML and DL models. Labeling a dataset means mastering the techniques to identify whether an account is a bot or not (Orabi et al., 2020). The more the data labels are accurate, the more the model is efficient. Generally, this problem needs to be studied more because most datasets researchers use are labeled. Manual labeling is still the most popular technique; it can be carried out by experts, crowdsourcing, or volunteers (Almerekhi and Elsayed, 2015). This process is time-consuming and inefficient due to the annotation errors that may occur, and only a small set of data can be produced. The second annotation technique is based on the list of suspended accounts (Morstatter et al., 2016), which were detected as bots by checking the status of the account (active,

deleted, suspended). The third method is based on honeypots, a group of accounts that behave like bots without interacting with normal users. This is a trap to attract social bots. Thus, any user interacting with this honeypot network will be considered a bot. This technique collects a large set of active bots with a certain reliability. Another annotation technique uses semi-automatic labeling (Teljstedt et al., 2015) by proposing predictive models to aid the labeler. And finally, automatic labeling (Bruno et al., 2022) is based on bot detection tools and APIs like the Botometer API (Yang et al., 2022). Table 15 presents the strengths and limits of each labeling technique used in the literature reviews and the references to the studies that used these techniques.

Unfortunately, there needs to be a clear definition of SMBs, which makes it challenging to develop a strategy for stabilizing the labeling process. Here are some behavioral patterns that are highly reliable for discriminating between social bots and normal users: 1) Absence of original and appropriate content and duplicate contents and posts several times; 2) Retweets are much higher than regular tweets from a genuine account; 3) Retweeting a vast number of tweets in a brief period; 4) The abundant presence of spam or illicit URLs in the posted content that redirects to phishing, malware, or adversaries' links. 5) Display links that do not match the post context; 6) The OSN accounts have a recent creation date; 7) User names and screen names include digits, which might point to an auto-generation of names; 8) Many users publish the same content at the same time, which refers to a synchronization activity launched by the botnet master; 9) Abusively utilizing the trending hashtags to advertise content irrelevant to the context of the used hashtags to gain more visibility and influence.

## 5. Discussion

This section discusses the results and conclusions reached by the SLR study and gives directions for future research. The main objectives of this paper were to summarize, analyze, and synthesize all the significant elements involved in the social bot detection process, including data source selection, data labeling, feature selection and extraction, and model training, as well as to highlight the limitations of the reviewed studies and to propose openings for improving the quality of SMBs detection systems.

**RQ1:** From the studied articles, we identified eight types of social profiles that operate in OSN. SMBs include three types: benign, neutral, and malicious. For benign SMBs, there is a category of social bots: promoters, celebrities, informational bots, and chat-bots. As well, for malicious SMBs, we have identified six types of social bots: 1) SMBs; 2) spam bots; 3) sybils and cyborgs; 4) political bots; 5) stegobots; 6) gambling bots, as shown in Fig. 3. The new generation of SMBs is provided by cloud services such as Bot-as-a-Service (BaaS), and these SMBs include intelligent functions such as sophisticated content generation similar to humans. Every day, new types of bots may appear that the present systems may not detect because they ignore their behavior. That is why it is necessary to follow their evolution and explore the different characteristics of an OSN account to detect them efficiently. It would be interesting to apply the different detection systems to all current SMBs to detect gaps and predict the evolution of the behavior of social robots over time. Studying emerging social media platforms like TikTok can help discover new types of social media bots. As new platforms emerge, they often introduce new features, technologies, and ways of engaging with users, creating new opportunities for bots to manipulate, influence, and deceive users. TikTok's video-based format and users' short attention spans provide a fertile ground for bots to generate automated videos that mimic human behavior and interactions.

**RQ2:** ML-based techniques are still the most used for bot detection, followed by DL techniques. The RF, SVM, and AdaBoost are the most employed algorithms for SMB detection, as illustrated in Fig. 7. The DT and RF achieved a precision score equal to 100% with models trained on the Cresci-17 dataset. Therefore, for the ML algorithm, RF was the most used algorithm for spam bots, which achieved an F-measure of 97.9%. The GB algorithm reached an accuracy of 98.97%. DL-based techniques need to be explored more, and the most commonly used DL algorithms are LSTM and CNN. Both the LSTM and CNN algorithms achieved a good precision of 99.8% and an accuracy of 100% (see Fig. 8 and Table 4). However, there is a lack of large datasets to train DL models, which can limit their use. It's crucial to involve social network users in the detection and prevention process of fake accounts through voting or trust techniques, as proposed in some articles. Integrating feature-based and graph-based techniques for fraudulent account identification has also received little attention. The proposed methods are still in their early stages and must be optimized. The supervised techniques are considered early detection, based on the principle of training a model on a dataset of OSN accounts labeled as human or malicious SMBs. The limit of these detection techniques is that evolving SMBs can escape the detection systems because social bots keep developing their characteristics and behaviors over time. Nowadays, social bots use advanced strategies to mimic human behavior by increasing their connections with real friends and followers and owning accounts with cloned names and profile pictures of real accounts to deceive OSN users and detection systems. However, the unsupervised techniques cluster the group of coordinated SMBs based on their synchronized behaviors and activities, which become inaccurate faces of the small groups of poorly coordinated bots that may need more information for their detection, which can lead to a biased detection system.

**RQ3:** It is evident throughout Table 4, Table 5, and Table 6 that the most commonly used features for specific or generic SMB detection were content, user information, and network features. The sentiment features effectively detect massively synchronized activities associated with elections or political conflicts. Therefore, behavioral features are typically used to detect Sybil, cyborgs, and game bots. To combat malicious SMB evasion, many researchers have used a variety of features to distinguish legitimate accounts from malicious ones. As we have already mentioned, the new generation of SMBs are provided with intelligent functions that allow them to generate and post textual content similar to human-generated text. Our study shows that contextual and deep-linguistic features need to be explored more by researchers to analyze textual content and posts, especially word embedding techniques and deep-NLP. Indeed, for future works, it will be very interesting to experiment with the impact of DL on NLP on the effectiveness of SMB detection systems. Therefore, social bots may exhibit different behaviors, so exploring additional features and models is essential to find the best approach. So, no single feature can accurately identify a social bot, and it's often necessary to use a combination of features and advanced machine learning algorithms to determine the relevant features accurately. Additionally, social bots are constantly evolving and adapting to evade detection, so it's essential to update and refine bot detection algorithms regularly.

**RQ4:** Finding accurate features is critical to build an effective system to detect SMBs. The feature engineering and preprocessing techniques in the social bot's detection field are discussed in many works, such as data collection, cleaning data, normalization, feature extraction, and selection. As depicted in Tables 8 and 9, feature extraction and feature selection techniques are not used extensively and sufficiently to detect SMBs effectively. Preprocessing social media data for bot detection while preserving the semantic meaning of emojis and the sentiment of the content can be chal-

lenging. Several strategies are to consider, such as tokenization, emoji-to-text conversion, word embeddings, and preprocessing pipelines. It is important to experiment with different techniques and pipelines to find the best way to prepare data for the training phase without resulting in a loss of information; some removed symbols or characters can be discriminant for the detection of SMBs. It would be interesting to start by extracting statistical features before cleaning the data, for example, calculating the number of URLs, number of motions, and number of hashtags, replacing emojis and emoticons with their meaningful words, and replacing acronyms with their complete forms. Additionally, it is crucial to analyze the data and adjust the preprocessing accordingly for each type of SMB. It is observable from Table 8 that the majority of work used the traditional bag of words techniques for feature extraction, which cannot capture the semantic and syntactic relationships between terms. The word-embedding has been shown to improve the performance of various NLP tasks, such as sentiment analysis and text classification. By capturing the semantic and syntactic relationships between words, word embedding provides a more nuanced understanding of language that can improve the accuracy of NLP models. Using word embedding can refine and improve detection methods. Based on the works we studied, we found that user information, content, and network features are the most used for SMB detection. While choosing features, accuracy, and speed must be balanced, dimensionality reduction is employed to condense the feature space while taking key features into OSN accounts. It's worth noting that no single feature can identify all bot accounts accurately. Rather, multiple features and techniques should be combined to create a robust bot detection system. Researchers are developing methods to identify each type of SMB separately. However, developing a detection system that can identify all SMB types using a single generic model with the same or comparable features would be interesting.

**RQ5:** Although the literature has already provided many public datasets for SMB detection. Unfortunately, most represent two generic classes, such as “legitimate” or “fake,” or focus on a specific type of SMB. So, they cannot be generalized to all SMBs. In addition, all these datasets depend on the Twittersphere and include limited data that does not allow us to recognize SMBs that exhibit advanced malicious behaviors. The majority of them provide only user information and content features. However, TwiBot-20 is the only dataset that can be used for community-based detection because it provides data on users' friends, followers, and content features in addition to user information and content features. Botometer is a tool developed by researchers to determine the likelihood that a given Twitter account is a bot or a normal account. It analyzes various account characteristics, including its tweeting behavior, metadata, and the content of the tweets. Researchers can use this tool to label Twitter datasets automatically. It also offers a repository of all the public datasets that the researchers provide. In Table 12, we listed all the public datasets provided by researchers for academic works, with detailed descriptions of the types of SMBs and the sizes of the datasets. On the other hand, in some works, the authors used web scraping techniques to study specific cases of SMB detection, such as political and spam bots. Indeed, data was extracted from OSNs based on hashtags, trending topics, or keywords. Therefore, to detect spam bots, fake accounts, Sybils, and cyborgs, most of the researchers used public SMB datasets. From Table 12 and Fig. 12, we can confirm that the most used public datasets by researchers are Cresci-17 and Cresci-2015. Furthermore, for spam bots, with the Cresci-17 dataset, ML and DL models produced good results with a precision that exceeded 95%. Hence, to identify fake accounts. The Cresci-2015 dataset, the most frequently utilized by researchers in various papers, has a precision of over 95%. Hence, Table 5 presents more information about the Cresci-2015 and Cresci-17 datasets. Social media plat-

forms often have restrictions on the data that can be accessed, limiting the data available for bot detection. Additionally, some social bots may use techniques to obscure their activity, making it difficult to detect them. To overcome the limited data availability, missing or confusing ground truth, and obsolescence of existing datasets, researchers must establish reference datasets that contain harmful accounts, such as social bots, cyborgs, and political trolls. The data availability problem requires the most attention from the scientific community and remains largely unsolved. Additionally, the data used for bot detection may be biased toward certain types of social bots, making it difficult to detect others. For example, bots that mimic human behavior may be more difficult to detect than those that exhibit anomalous behavior. Therefore, imbalanced datasets are a common challenge in social media bot detection. In many cases, there may be significantly more data for one class (e.g., human users) than the other (e.g., spambots). Imbalanced datasets in social media bot detection can lead to poor performance for the minority class. Researchers applied many different techniques to handle the issue of data set size and biased class proportions, such as random oversampling, SMOTE, CGAN, SMOTE, and ADASYN (see Table 14). To address this, strategies such as resampling, synthetic data generation, and data augmentation can be used to increase the number of samples and improve the model's performance. However, there is no one-size-fits-all solution for imbalanced datasets, and the optimal approach will depend on the dataset and problem. The issue of dataset labeling for social media bot detection is a challenging problem that requires careful consideration and expertise. The challenge with dataset labeling is that it can be difficult to distinguish between human users and bot accounts. This is because bots can be designed to mimic human behavior, such as tweeting at certain times of the day or using natural language. Additionally, bot detection algorithms can be susceptible to bias, leading to inaccurate results. Self-supervised learning is an improvement over unsupervised learning methods that learn from input data to learn data representations to solve data labeling problems automatically, even for large datasets. Some of the strengths of this technique are its scalability and reduction of time costs by limiting human intervention. However, semi-supervised learning uses both labeled and unlabeled data to train the model, and using these techniques will improve the labeling process and reduce the cost of time. Moreover, we noticed that no public datasets are addressed for stegobots detection. Stegobots use steganographic images to conduct their attacks. So, rather than using only standard features, it would be interesting to investigate how GANs can help to provide realistic stereographic images to refine SMB identification.

## 6. Conclusion and perspectives

Considering the dangerous impact that malicious SMBs play on OSNs, this has led researchers to dig deeper into this line of research and propose detection systems that help limit their invasion. But this has not stopped the creation of newly evolved SMBs and techniques that can bypass the detection systems. This state-of-the-art aims to identify existing SMB detection systems by applying SLR instructions and guidelines. We reviewed and assessed previous research works in the field of SMBs published in research databases such as ScienceDirect, IEEE, SpringerLink, and Google Scholar. Given the large number of published works that deal with the problem of social bots' detection, we studied only the articles published between 2008 and 2022; we also excluded works that have a small number of citations and those that have not been indexed, which can be considered the limit of this work. We used the PRISMA approach to conduct a bibliographical search of our SLR; nevertheless, only 144 pertinent pub-



lications were found, exposing the stages of our search strategy. We first identify the research questions relevant to our study, analyze the relevance of the recovered articles using inclusion and exclusion criteria, and then evaluate the works gathered using quality evaluation methods before attempting to answer the research questions we initially identified. Based on the research retrieved, we classified OSN profiles into three categories: genuine, verified, and cloned or false accounts. We classified social bots into three main groups: benign bots, which automatically post alert tweets, distribute news, engage with users, and attend to their needs; neutral bots, which post or repost jokes; and malicious bots, which are directed by a bot master to carry out automatic tasks and plan attacks. SMBs, spam bots, Sybil and cyborgs, stegobots, political bots, and social game bots are six categories of malicious bots. We conducted related research on each sort of malicious bot based on this classification to complete a pertinent synthesis and analysis of the examined methodologies. We proposed a taxonomy of the SMB's detection technique, which includes eight categories: ML-based, DL-based, graph-based, anomaly-based, topic modeling-based, DNA-inspired, genetic-based, and hybrid approaches. Then, we examined these techniques and showed the strengths and weaknesses of each studied approach by systematizing, evaluating, incorporating, and presenting the outcomes and findings of the reviewed works. The results of our study show that it is so challenging to implement efficient SMB detection systems because social bots keep changing their characteristics over time, and the new generation of SMBs have intelligent functions such as sophisticated content generation similar to humans. To detect them efficiently, it is necessary to track their evolution and investigate the various characteristics of an OSN account. It would be an ideal scenario for the near future. It would also be useful to compare detector performance to that of existing SMBs to simulate how bot characteristics change over time. The RF, SVM, and AdaBoost are the most used ML algorithms for SMBs detection. The most commonly used DL algorithms are LSTM and CNN. However, there is a lack of large datasets to train DL models. Early detection is based on training a model on a dataset of OSN accounts labeled as human or malicious SMBs. Still, these techniques' limits are that evolving SMBs can escape the detection systems. Self-supervised learning is an improvement over unsupervised learning methods that learn from input data to learn data representations to solve data labeling problems automatically, even for large datasets. Existing approaches address these issues, but further enhancements may still be required. There is also a need to implement models that detect different types of SMBs, even those outside the Twittersphere; moreover, certain methods need further exploration, such as leveraging NLP techniques or anomaly-based methods. The most commonly used attributes for conducting specific or generic SMB detection were content, user information, and network characteristics. To combat malicious SMB evasion, researchers have used a variety of features to distinguish legitimate accounts from malicious ones. However, the researchers did not sufficiently explore contextual and deep-linguistic features to analyze textual content deeply, especially word embedding techniques. Deepfakes, an AI-enabled deception method, can now blur the lines between legitimate accounts and the most sophisticated malicious accounts. Feature extraction is essential for ML algorithms to recognize social bots accurately, but no single feature is enough to identify all accounts accurately. Researchers must focus on identifying robust and differentiating traits to create a robust bot detection system. Many public datasets for SMB detection have been published in the literature. Still, most represent two generic classes or are focused on a specific type of SMB. The only dataset that can be used for community-based detection is TwiBot-20. To overcome the limitations of limited

data availability, missing or ambiguous ground truth, and the obsolescence of existing datasets, researchers must create reference datasets containing harmful accounts and devise new methods for producing adversarial cases. These issues are largely unsolved and require the most attention from the scientific community. Therefore, the progress of science depends on the policies adopted by decision-makers and operators of OSNs, because if researchers have advanced access to more data, they will be able to respond to new developments and challenges in this field.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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